

'The democratic touch of disasters': The perception of non-governmental organizations in public institutions in the context of disaster governance

Oğuzhan Özkan^{*}, Alper Uzun

Department of Geography, Balıkesir University, Balıkesir, Türkiye

ARTICLE INFO

Keywords:

Balıkesir disaster governance
NGOs
State
Türkiye

ABSTRACT

As in all matters related to society, governments have the last word in disaster management. But the views of public institutions and state-affiliated organizations are also crucial. This study examines the perspectives of public institutions in Balıkesir, reflecting Türkiye's disaster sensitivity, on the role of non-governmental organizations (NGOs) in disaster management. The aim is to assess the functionality of disaster management and governance in Balıkesir. Semi-structured interviews were conducted with 9 public institutions with significant disaster responsibilities, and the data were coded and analyzed using MAXQDA 2020. While public institutions acknowledge that NGOs provide valuable labor during disasters, they generally believe that disaster management would continue without them. However, the February 6, 2023 Kahramanmaraş earthquakes challenged this view, as international aid, including from NGOs, became essential. The Disaster and Emergency Management Authority (AFAD) representative differs from this dominant perspective, emphasizing the critical role of even small NGO contributions. Countries like Japan and the United States of America (USA) highlight the important role of NGOs in disaster management. For instance, Voluntary Organizations Active in Disaster (VOAD) in the USA is a key partner in local disaster response, benefiting from its initiative in post-disaster activities. Conversely, Turkish public institutions often view AFAD as superior and criticize NGOs for lacking leadership and order. This suggests that both the Turkish Disaster Response Plan's (TAMP) implementation and the internal and external challenges faced by NGOs reveal that disaster governance in Balıkesir is not functioning optimally.

1. Introduction

The struggle of the world societies against disasters and the processes of disaster reduction are directly related to sustainable development. Moreover, there is an absolute mutual dependence between sustainable development and combating disasters. One of the indicators of this interdependence is that disasters interrupt the success of sustainable development. This is because disasters can seriously threaten and disrupt social, political, economic and spatial order. Deficiencies in the disaster management systems built by societies can also mean a great risk for possible disasters [1].

The inextricable link between sustainable development and disasters constitutes one of the most important reasons why disaster

^{*} Corresponding author.

E-mail addresses: oguzhanozkan23@gmail.com (O. Özkan), aunuzun@balikesir.edu.tr (A. Uzun).

management systems should be based on sustainability as well as development. An unsustainable disaster management is likely to exhibit an inefficient and unsuccessful profile that does not address every period. One of the most important building blocks of sustainable disaster management is to ensure social participation. The more all segments of society take part in the disaster management mechanism, in other words, the more they participate in the system and become partners, the more sustainable disaster management becomes. This situation essentially reveals the necessity for the disaster management organization to focus on society. Because, as Palttala et al. [2] state, due to the increasing complexity of disasters, it is essential that organizations representing society, other than governments, contribute to disaster management activities. Therefore, the existence of an environment that will ensure social participation from the planning stage of disaster management activities is of utmost importance [1,3].

In emergencies and disasters, volunteerism - in other words, civil society - is of great importance. Dostál [4] also interprets this point in terms of emergencies and disasters being indispensable subjects of public security policy and establishes a cause-and-effect relationship. According to the author, due to the value of civil society, financial support for emergency and disaster-oriented non-governmental organizations (NGOs) is mostly provided from the budget allocated for disaster management. In addition, when necessary, civil society can also fill the gaps created by governments. For example, in the aftermath of the 1974 floods in Bangladesh, the government of the time was completely ineffective in providing material assistance to the disaster victims. Therefore, in the eyes of the people, the government became a failed and even corrupt organization. Thus, humanitarian aid in Bangladesh was mostly provided through NGOs. The same was the case in the Czech Republic after the devastating floods of 1997, when NGOs were more practical and successful than public institutions. Thanks to their success, NGOs gained a high status and trust from the public [4].

In Türkiye, which is an extremely disaster-sensitive country, the rise of civil society in disaster management took place after the 7.4 earthquakes centered in Gölçük on August 17, 1999 and the 7.2 earthquakes centered in Düzce on November 12, 1999. Just like in the cases of Bangladesh and the Czech Republic, it is understood that public institutions were ineffective after these disasters in Türkiye. This is because there are major shortcomings in terms of pulling disaster victims out from under the rubble, meeting their needs and providing other assistance. These deficiencies were addressed by the Search and Rescue Association (AKUT), a disaster-themed NGO. Thanks to AKUT's rescue of many people from under the rubble and the delivery of aid, society's view of civil society has changed positively. On the other hand, the impact of the change could not be deep-rooted enough in the disaster management organization in Türkiye. Because the problems witnessed in various disasters after this period and most recently in the 7.7 and 7.6 magnitude earthquakes centered in Kahramanmaraş on February 6, 2023 reflect the inadequacy of the power and value of civil society. In the aftermath of the February 6, 2023 Kahramanmaraş earthquakes, the combined resources of public institutions and national NGOs were not enough to cope with the disaster, and Türkiye was forced to request international assistance. 82 countries, including the USA, France, Greece, Armenia, Greece, Armenia and Israel, which sometimes had diplomatic disputes with Türkiye, accepted the request and sent more than 9 thousand public and NGO personnel with expertise in important areas such as search and rescue, humanitarian aid and setting up field hospitals to the disaster areas [5,6]. Despite all these efforts, 53 thousand 537 people lost their lives, and 107 thousand 213 people were injured due to the earthquake [7].

When examining the relevant literature, it is understood that studies primarily focus on the roles of civil society in disaster management [8–15], the connections between public institutions and civil society [16–19], societal perceptions of the importance of NGOs in disaster management [20–22], and the perceptions of vulnerability in disaster management by public institutions and civil society [23].

As in all matters concerning society, in disaster management, governments are the ultimate decision-makers. The perspectives of public institutions or organizations operating within the state mechanism implemented by governments regarding NGOs are therefore of significant importance. It is observed that there are limited studies in Türkiye that comprehensively examine the roles of NGOs in disaster management and the perspectives of public institutions on NGOs in this context.

Türkiye is a country highly exposed to both natural and human-induced disasters. Natural events such as earthquakes, floods, landslides, rockfalls, avalanches, erosion, forest fires, and droughts frequently turn into disasters, causing significant loss of life and property [24,25]. According to EM-DAT [25] data, 211 natural disasters occurred in Türkiye between 1900 and 2025, and in 2023, Türkiye ranked 8th among the countries experiencing the most natural disasters worldwide [26]. Notably, 114 of these 211 disasters were earthquakes, while 50 were floods. Another critical point is that the vast majority of casualties resulting from natural disasters are caused by earthquakes. Over the past 125 years, natural disasters have claimed the lives of 150,114 people, with 147,021 of these deaths attributed to earthquakes [25].

Due to its geologically young formation, Türkiye is home to numerous active fault lines, making it one of the world's high-risk earthquake regions. As a result, Türkiye experiences major earthquakes causing significant loss of life and property almost every five years [24,27]. The two earthquakes that struck Kahramanmaraş on February 6, 2023, served as a tragic reminder of Türkiye's earthquake hazard. Additionally, the anticipated major earthquake in İstanbul is another crucial concern. As Türkiye's largest city and the country's cultural, economic, and financial hub, İstanbul is vital to the nation's well-being. Given its population nearing 20 million, an earthquake in İstanbul would mean a catastrophic disaster for Türkiye.

Regarding human-induced disasters, 176 incidents occurred in Türkiye between 1900 and 2025, with 71 of them resulting from road accidents and 36 from maritime accidents. Over this 125-year period, human-induced disasters have claimed 7053 lives [25]. Moreover, according to the Inform Report Index [27], Türkiye has been among the countries with an increasing risk of human-induced disasters between 2015 and 2024, a point that warrants particular attention.

This study investigates the views of public institutions and organizations located in Balıkesir, a province that can be considered a reflection of Türkiye in terms of disaster sensitivity due to its location on an active fault line and its exposure to disasters such as floods, landslides, wildfires, rockfalls, and hailstorms. The study aims to answer the questions of how the relationship between public institutions and NGOs in disaster management functions in Balıkesir and whether NGOs have a say in decision-making processes. In

doing so, the research seeks to evaluate the functionality of disaster management and the presence of disaster governance in Balıkesir.

In the introduction of the study, the relationship between sustainable development and efficient disaster management systems, as well as the importance of NGOs in these two dimensions, is briefly discussed. Subsequently, the conceptual framework of the study is presented, explaining the disaster risk reduction model and disaster (risk) governance, which constitute the key concepts of the topic. To highlight the position and significance of NGOs in disaster management, evaluations related to some developed and developing countries are provided. In the methodology section, the research population, sample, data collection, and analysis processes are specified. Under the findings section, the identified codes, sub-themes, and key findings emerging from the analysis process are discussed. Finally, in the conclusion and discussion section, the findings are interpreted in relation to the existing literature.

2. Background

2.1. Disaster risk reduction (DRR) model

Various risk factors that may threaten the survival and existence of societies can be mentioned. Naturally, societies resort to certain measures in order to completely prevent the relevant factors or reduce their effects. Precautions also require a preparation process and structuring. Because, the measures taken have the potential to be more efficient and successful if they are within a plan. Therefore, detailed and thoughtful planning of the preparation process is extremely important. Different theories and models have been developed in order to create this planning accurately. Some of the theories and models in question are; community engagement theory, faultline theory, social capital approach, social identity theory, critical awareness theory, person relative to event theory, protective action decision model and theory of planned behavior. The theories and approaches mentioned have a close relationship with each other, in other words, they have been influenced by each other. They have also shaped the preparedness policies -a kind of risk management-created against many risk factors [28].

Disasters are perhaps the most important risk factor that threatens societies. It is seen that different approaches are adopted to stand against disaster risk and that these approaches consist of or are influenced by the models and theories mentioned in the previous paragraph. One of these approaches is the Disaster Risk Reduction (DRR) model. DRR is the process of developing plans, policies, strategies and practices to reduce social fragility, build a resilient society and thus reduce disaster risk. This approach addresses a multitude of potential factors that could directly or indirectly threaten societies. These threats can present environmental, geological and meteorological characteristics as well as socioeconomic, cultural, political and technological ones. The factors in question, each of which is a disaster trigger, should be addressed in all aspects in DRR. In addition, it is seen that disasters are not perceived in DRR as events that need to be solved in the short term, but as deep-rooted and long-term processes that need to be examined meticulously and carefully. As a result of this temporal perspective, DRR also brings different advantages. In fact, effective and efficient implementation of DRR can create a suitable basis for reducing disaster vulnerability and increasing disaster resilience as well as achieving sustainable development [29].

As stated before, the planning, evaluation and implementation of DRR are not carried out depending on specific intervention types and time constraints but are carried out with a holistic and systematic approach. The holistic and systematic character of the approach also brings with it the direct relevance of social, economic, natural, environmental, health and engineering sciences and techniques to DRR. In this sense, one of the most important processes representing the multiple perspectives in DRR is the Hyogo Framework for Action (HFA) 2005–2015, which was approved at the Second World Conference on Disaster Reduction held in Kobe, Japan in January 2005. This plan, signed by UN member countries, is based on five basic articles:

- Establishing a strong institutional structure for the implementation of DRR,
- Establishing strong early warning systems as well as determining, analyzing and monitoring disaster risks,
- Preferring information, education and innovative approach in order to build a culture of security and resilience in every segment of society,
- Reducing disaster risk factors,
- Making disaster preparedness effective and strong [29],

Another plan that is very effective in terms of DRR, like HFA, is the Sendai Framework for Disaster Risk Reduction (SFDRR) 2015–2030. This plan, approved at the Third World Conference on Disaster Risk Reduction in March 2015, has four priority areas:

- Determining disaster risk,
- Strengthening disaster risk governance to manage disaster risk,
- Making investments to reduce disaster risk to ensure disaster resilience,
- Improving the disaster preparedness process for effective post-disaster intervention and aiming for better search and rescue, rehabilitation and reconstruction [29–33].

The concept of innovation is highly valued in DRR. Accordingly, as seen in SFDRR, investments in long-term and solution-oriented innovation and technological development are encouraged for DRR [34]. It is understood that innovation is addressed in the context of several subgroups:

- Innovation related to interdisciplinary concepts such as disaster resilience,

- Innovation to maintain GIS and make it more qualified with practical tools such as artificial intelligence,
- Innovative methods of scientific knowledge-based policy production and implementation,
- Community-based innovation such as incorporating local-scale information into DRR decision-making processes,
- Inclusive and participatory innovation that integrates the approaches of young scientists into DRR,
- Innovation that monitors the consistency and development of disaster policies [35].

Scientific knowledge production requires collaboration among scientists, civil society, policy makers and practitioners [35,36]. In this way, different perspectives are more effectively integrated into the joint assessment of experts and decision makers, and the opportunity to provide objective solutions to problems increases [36–38]. Moreover, Alexander [39] defends the idea that one of the most important and efficient methods in disaster management and disaster risk reduction is disaster governance. According to him, disaster governance is possible not with a top-down management approach, where governments dictate to other segments of society and give them little say, but with a bottom-up management approach, where all social stakeholders have as much say as governments in disaster policies. Therefore, the holistic approaches mentioned in DRR and the direct contribution of civil society to disaster risk reduction processes can be evaluated in the context of bottom-up disaster management.

2.2. Disaster (risk) governance

Disaster governance is a concept that has recently entered the literature as a potential method for reducing disaster risks [40] and is also one of the five main priorities adopted in the Hyogo Framework for Action (HFA) [41]. The concept refers to the combination of interlinked regulatory rules and frameworks, institutions/organizations, and practices for anticipating, combating, resisting and recovering from natural or man-made hazards within the disaster cycle (disaster response, search and rescue, reconstruction, mitigation and preparedness). It can also be organized at various social and spatial scales [42]. Tierney [43] also has a parallel idea with Gall et al. [42]. Tierney argues that disaster governance refers to a set of norms, institutions, organizations, and practices that are established to mitigate the effects of disasters caused by natural and technological factors as well as terrorist activities and any kind of loss. As in the previous definition of disaster governance, the sets created here cover the entire disaster cycle. According to Tierney, disaster governance is a system that motivates the participation and collaborative activities of all social stakeholders (government, private sector, NGOs and universities) that can occur at all geographical scales from local to global [43]. Due to the ultimate goal of disaster governance including pre-disaster, the concept is also used as "disaster risk governance" [42].

In disaster risk governance, it is of utmost importance to question the factors that create disasters. In this approach, disaster is not caused by God, nature or any other external factor, but by social or cultural deficiencies. In this approach, poor governance, vulnerabilities based on sociospatial structures, and social vulnerabilities are primarily blamed for all the losses and destruction that occur as a result of disasters [44–46]. As a matter of fact, with the presupposition that human factors are the reason why natural events take the dimension of disasters, it is argued that reducing or eliminating disaster risks also depends on people, and the main goal is to reduce social vulnerabilities and increase disaster resilience [21,47]. From this perspective, it is understood that disaster governance adopts a perspective opposite to the traditional disaster management approach and in parallel with modern disaster risk management.

There are studies on disaster vulnerability at different spatial scales. Internationally, there are studies on the impact of social inequalities [48–50] and intra-national social inequalities on disaster vulnerability. Intra-national studies mostly try to explain how the poor, disabled, children and migrants [51], those in need of special care [52], women [53–55] and the elderly [56–58] are more vulnerable to disasters than other segments of society. This vulnerability highlights the shortcomings in the disaster management systems of some countries and, consequently, in their response to disasters. The disadvantaged groups, who are the subjects of vulnerability, form a crucial part of civil society, and their marginalization in disaster management reflects the unhealthy nature of disaster governance. Therefore, relevant studies argue that there is a positive correlation between considering the demands of disadvantaged groups in disaster management processes and the construction of a democratic disaster management system.

In order to ensure disaster resilience, the root causes of disaster risks such as rapid urbanization, poverty, economic insecurity and food insecurity are tried to be solve. Therefore, disaster risk reduction is closely related to the reduction of vulnerability and is directly linked to development activities. In disaster governance, the fact that unsustainable development systems reduce disaster resilience, and the occurrence of disasters is explained by this relationship [59]. As mentioned earlier, integrating the community into the process is an important condition for establishing a sustainable disaster management approach, and the aim of increasing disaster resilience in this direction necessitates community-based disaster governance. Thus, considering the multi-actor nature of the governance concept, it can be stated that community-based disaster risk governance constitutes a model covering individuals, households, communities, private sector, universities, civil society, military organizations, donors, multilateral organizations and the government. In disaster governance, all these actors share responsibilities for disaster risk reduction [21,45].

2.3. The role of civil society in disaster governance

Civil society plays a crucial and multifaceted role in disaster governance worldwide, but the nature and effectiveness of this role vary significantly between developed and developing countries. In developing countries such as Bangladesh, Pakistan, Nepal, Nigeria, and Uganda, disaster management systems have historically focused on emergency relief and rehabilitation. Over time, these systems have gradually evolved to incorporate broader functions including disaster forecasting, preparedness, risk reduction, and reconstruction [15,16,60–62]. However, in many of these countries, civil society participation—especially during disaster preparedness and planning phases—is often limited due to centralized state control or political and cultural barriers.

Bangladesh has made notable strides by institutionalizing disaster management bodies such as the Disaster Management Bureau (DMB) and including diverse civil society actors through advisory committees, which involve NGOs, business leaders, and community representatives [60]. Despite this, challenges persist in ensuring meaningful and continuous civil society involvement, particularly at the grassroots level. In Pakistan, socio-political constraints and cultural norms marginalize vulnerable groups such as women, resulting in their limited participation not only as disaster victims but also as active agents in disaster preparedness and response [15,63]. Similarly, in Nepal, bureaucratic delays and poor coordination between government agencies, local governments, and NGOs hamper effective disaster governance, with community-based management strategies still struggling to take root [16,64].

In Nigeria, a country frequently exposed to both natural and human-made disasters, civil society has gained increasing recognition for its essential role in complementing state efforts. The establishment of the National Emergency Management Agency (NEMA) and inclusive policy frameworks like the 2014 Strategic Response Plan have fostered stronger cooperation among government bodies, national NGOs, and international partners [61,65]. Despite this progress, capacity gaps within both government and civil society actors remain a significant barrier, particularly in initial response phases, leading to initiatives such as the Emergency Management Volunteers (EMV) program to strengthen local disaster response capabilities [65]. Uganda faces similar challenges, with limited financial and human resources constraining disaster preparedness and response. However, Uganda's National Policy for Disaster Preparedness and Management emphasizes multi-stakeholder participation and inclusiveness, aiming to integrate vulnerable groups and NGOs into national disaster governance frameworks [62,66].

In contrast, developed countries exhibit more mature and institutionalized disaster governance systems where civil society engagement is deeply embedded and systematically supported. The United States offers a prominent example through the Federal Emergency Management Agency (FEMA), which actively coordinates with a wide range of civil society organizations—including community-based groups, NGOs, and volunteer networks—to implement comprehensive disaster preparedness, mitigation, and recovery programs. FEMA's strategies prioritize inclusivity by addressing the needs of marginalized populations such as the elderly, disabled, and low-income communities, ensuring that disaster risk management is equitable and effective [67–71].

Germany exemplifies a multi-level disaster governance model where federal, state, and local authorities collaborate closely with civil society actors such as the German Red Cross and numerous volunteer organizations. Public awareness campaigns, regular training, and simulation exercises are key components of Germany's approach, supported by robust legal frameworks and technological infrastructure [72]. These measures contribute to a well-coordinated response system that integrates civil society into every stage of disaster management.

Sweden adopts an "integrated" and "community-based" approach to disaster management, emphasizing extensive cooperation with civil society organizations. According to the Swedish Civil Contingencies Agency (MSB), preparedness and effective disaster planning require the involvement of all societal stakeholders [73–75]. A notable example of this collaboration is MSB's partnership with the Swedish Red Cross and the Swedish Civil Defense Association. In 2019, MSB designated the Red Cross as the leading organization for informal volunteering, marking a critical step in incorporating informal volunteers into the national disaster management system [73]. Similarly, in Finland, the role of volunteers is formally outlined under the Rescue Act. The Volunteer Rescue Service (Vapepa), a network of 53 associations such as the Finnish Divers' Federation and the Lifesaving Institution, coordinates volunteer activities, both upon government request and through its own initiatives, recognizing volunteers as key stakeholders even in large-scale emergencies [12].

Norway and Australia present different but equally impactful models. In Norway, public institutions—especially local governments—maintain positive relationships with NGOs. However, informal volunteers are seen as short-term partners, mobilized primarily in post-disaster contexts. To support their involvement, short-term contracts are established to provide insurance and compensation in case of injury or death during response efforts [75–77]. In contrast, Australia regards civil society as a long-term and sustainable means of fostering disaster resilience. A localized, community-based, and flexible approach is applied, involving both governmental institutions and NGOs. Moreover, the initiatives of households and local communities—viewed as integral components of broader civil society—are considered crucial in strengthening stakeholder cooperation within the disaster management system [10, 19,78,79].

Japan stands out globally for its advanced disaster risk management system shaped by frequent exposure to earthquakes and tsunamis. The country combines cutting-edge technology, such as early warning systems and GIS-based risk assessments, with a strong culture of community participation. Local volunteer groups, NGOs, and government agencies work hand-in-hand to conduct extensive education programs, risk mitigation efforts, and rapid response operations. This integration of technology and societal engagement enhances Japan's resilience and serves as a global benchmark [13,71,80–84].

The comparison between developing and developed countries highlights key differences in civil society's role in disaster governance. Developed countries tend to have well-established legal frameworks, sufficient resources, and inclusive policies that facilitate effective multi-stakeholder collaboration. Civil society organizations in these contexts possess greater capacity and autonomy, allowing them to contribute proactively across all disaster management phases. Meanwhile, developing countries often grapple with limited resources, centralized governance structures, socio-cultural barriers, and capacity constraints that hinder civil society's full participation.

Nonetheless, there are clear pathways for developing countries to strengthen their disaster governance by learning from the best practices of developed nations. Emphasizing decentralized governance, capacity building for civil society, inclusive policies that recognize marginalized groups, and leveraging technology for disaster preparedness can gradually transform disaster management systems. The evolution observed in countries like Bangladesh and Nigeria indicates growing recognition of civil society's indispensable role, which—if further supported—can enhance resilience and reduce disaster risks effectively.

2.3.1. Civil society's role in Türkiye's disaster management: shifts in governance and collaboration

The two major earthquakes that struck Türkiye on August 17 (centered in Gölçük) and November 12 (centered in Düzce) in 1999 were turning points for the country, including in the field of disaster management. The most significant outcome of these turning points was the society-based mental change and transformation. Until these devastating earthquakes, the standard perception of disaster management and the relationship between the state and civil society in Türkiye was altered, leading politicians, scientists, and journalists to introduce new discourses regarding these relations [21]. Akgüngör [85] describes this mental transformation, particularly for Turkish intellectuals, as a “*milestone*”, a characterization that has been accepted by many other researchers [54,80,86,87]. In his study, Akgüngör [85] argues that the state was highly ineffective and unsuccessful during the earthquakes, leading to the erosion of the “*all-powerful*” and “*fatherly*” image of the state. He further states that this loss of confidence paved the way for the rise of AKUT, one of Türkiye's most prominent search and rescue organizations. It must be acknowledged that poorly managed disaster processes often lead to structural reforms [88–90]. This was also the case in Türkiye, where the inadequacies exposed by these disasters prompted the elimination of the multi-headed structure in disaster management. Consequently, in 2009, Türkiye established the Disaster and Emergency Management Authority (AFAD), the highest authority in the country's disaster management system.

Regarding disaster governance, some researchers in the literature argue that Türkiye's position and level of development remain inadequate. Özerdem and Jacoby [80] discuss the centralized state structure in Türkiye, emphasizing that the political elite do not tolerate civil society within the disaster management system and impose heavy restrictions on NGOs. According to the authors, this situation is directly linked to concerns within the state that NGOs might use humanitarian aid as a political tool. Both Jalali [86] and Özerdem & Jacoby [80] argue that state-imposed pressure on civil society is the main factor behind its weakness, thereby attributing the lack of progress in disaster governance to state policies. Additionally, Aydın and Özgür [91] assert that the imbalance between a strong and dominant state mechanism and a weak civil society leads to low performance in disaster management organizations. According to these authors, this situation creates security vulnerabilities in the face of disasters.

Despite the challenging and controversial environment surrounding the role of NGOs in disaster management, state policies have assigned significant responsibilities to NGOs. A notable example of this is the Türkiye Disaster Response Plan (TAMP). Adopted by the High Council for Disaster and Emergency Management on December 20, 2013, through Decision No. 2013/2, and published in the Official Gazette No. 28871 on January 3, 2014, TAMP aims to establish the principles of disaster response planning before, during, and after a disaster. Another crucial aspect of the plan is the definition of the roles and responsibilities of service groups and coordination units tasked with disaster and emergency response activities. TAMP emphasizes the significant role of NGOs as stakeholders within the framework of the plan. Additionally, it states that NGO representatives should be included in the Provincial Disaster and Emergency Coordination Board, which operates in every province under the leadership of the provincial governors [92]. However, it is important to note that while the plan recognizes the role of NGOs and represents a development in collaborative disaster management in Türkiye, this collaboration is more evident at the national level than at the local level [17].

NGOs also serve as important support partners within fifteen critical working groups that play key roles in disaster and emergency situations. In this capacity, they assist in coordinating the efforts of ministries, institutions, and organizations that act as the primary solution partners. These working groups cover a wide range of functions, including disaster communication, search and rescue, health services, evacuation and planning, transportation and its infrastructure, shelter provision, nutrition, psychosocial support, debris removal, agriculture and livestock management, victim identification and burial, management of in-kind donations and warehouse distribution, technical support and supply, as well as national and international cash donation services [92].

3. Method and approach

3.1. Sample and data collection

The universe encompassing the public institutions that constitute the subject of this study includes disaster organizations (disaster coordination center, disaster commissions, fire departments) within the metropolitan and district municipalities of Balıkesir province, emergency, disaster, and civil defense units in state hospitals, disaster units within the provincial and district health directorates, as

Table 1
Sampled public institutions and the districts where they are located [108].

Name of public institution	District located
The Disaster and Emergency Management Authority (AFAD) Balıkesir	Karesi
BBB Fire and Disaster Management	Altieylül
Balıkesir State Hospital's Emergency and Disaster Unit	Karesi
The Occupational Health and Safety Unit of the Balıkesir Provincial Directorate of National Education	Altieylül
The Balıkesir Provincial Health Directorate's Health Services in Disasters Unit	Altieylül
Balıkesir Provincial Health Directorate's Health Disaster Coordination Center (SAKOM)	Altieylül
The Civil Defense Unit of the Balıkesir Provincial Directorate of Agriculture and Forestry	Karesi
The Forest Fire Fighting Branch of the Balıkesir Regional Forestry Directorate	Karesi
Balıkesir UMKE	Altieylül
Bandırma Training and Research Hospital's Emergency and Disaster Management Unit	Bandırma
The Burhaniye Municipality Earthquake and Natural Disaster Commission	Burhaniye
The Edremit Municipality Disaster Affairs Commission	Edremit
Susurluk State Hospital's Emergency and Disaster Management Unit	Susurluk

well as AFAD. Due to the large number of these institutions, the necessity of sampling has arisen. In line with the purpose of the study, purposive sampling was preferred by prioritizing institutions with greater responsibilities in disaster management. The selection of the institutions included in the sample was based on the guidance of a prominent representative from AFAD's Balıkesir branch, which is the top institution in Türkiye's disaster management system. Accordingly, the determined sample is presented in Table 1, while the map reflecting the research universe and the sample of public institutions is shown in Fig. 1. Within the scope of the research, interviews were conducted with nine institutions, as the Balıkesir Provincial Health Directorate's Health Services in Disasters Unit, Bandırma Training and Research Hospital's Emergency and Disaster Management Unit, Balıkesir Provincial Health Directorate's Health Disaster Coordination Center (SAKOM), and Susurluk State Hospital's Emergency and Disaster Management Unit declined to participate. Another reason for limiting the interviews to this number is that these institutions are the leading organizations in Balıkesir's disaster management. A final reason is the realization that participants' responses to the questions were becoming repetitive, in other words, reaching data saturation.

Among the public institutions included in the sample, three in particular hold significant importance for both Balıkesir and Türkiye's disaster management. To provide examples from recent large-scale disasters, AFAD Balıkesir, BBB Fire and Disaster Management, and Balıkesir National Medical Rescue Team Unit (UMKE) played a crucial role in responding to the forest fires that affected multiple provinces, including Antalya, Adana, and Muğla, in 2021, as well as the search and rescue operations following the Kahramanmaraş earthquakes on February 6, 2023, by deploying a large number of personnel. Additionally, these institutions provide disaster awareness training to other public institutions and students at all educational levels. Similarly, the Occupational Health and Safety Unit of the Balıkesir Provincial Directorate of National Education and the Civil Defense Unit of the Balıkesir Provincial Directorate of Agriculture and Forestry offer educational services to public institutions and students to raise disaster awareness. The Forest Fire Fighting Branch of the Balıkesir Regional Forestry Directorate prepares annual fire action plans and fire assessment reports, sharing them with the BBB Fire and Disaster Management. Furthermore, this institution collaborates with and provides technical support to various stakeholders, including schools, universities, forest villagers, and the Turkish Armed Forces, in areas related to the prevention, extinguishing, and aftermath management of forest fires. Most recently, the institution was involved in responding to the forest fire that broke out in Sındırgı on March 23, 2025. Lastly, the Burhaniye Municipality Earthquake and Natural Disaster Commission and the Edremit Municipality Disaster Affairs Commission prepare emergency action plans and work programs, particularly concerning earthquakes and other natural disasters. These plans and programs are periodically shared with AFAD. Additionally, the commissions undertake significant responsibilities in disaster management by working to identify disaster gathering and shelter areas in Burhaniye and Edremit districts. Ultimately, an important point to note is that these nine public institutions conduct comprehensive disaster drills within the Balıkesir province every year.

Semi-structured in-depth (open-ended) interview method was used to obtain data. This method allows for detailed and multidimensional evaluations on the researched topic. The questions in the interviews were grouped under the headings of institutional background and member profile, relations with NGOs and perceptions on the position of NGOs in disaster management. Various sources were examined in determining the interview questions. Questions regarding public trust in NGOs' capacity and role in disaster management, the relationship between NGOs and public institutions, and the potential of public institutions and NGOs in assisting the public during a disaster were derived from Çakı's [21] study. The question about the roles of NGOs in different stages of disaster management was based on the study by Erkal and Değerliyurt [93]. Questions on whether NGOs play a role in post-disaster recovery processes and whether they contribute to public disaster preparedness were drawn from the research of Benson, Twigg, and Myers [94]. Lastly, the question of how effective NGOs would be in the absence of the state and vice versa in disaster management was inspired by Carter's [95] study. As discussed in previous sections, social participation is crucial for sustainable development and, consequently, for building sustainable disaster management. One of the most significant indicators of social participation is the presence of NGOs within the system. Therefore, the study carefully formulated questions aimed at understanding the perspectives of decision-makers on the role of NGOs in disaster management, the impact of NGOs on sustainable development, and the state-NGO relationship. The prepared questions underwent a pilot study, first being reviewed by two academics specializing in disaster management and subsequently by a representative from AFAD Balıkesir. During this stage, the questions were found to be clear, comprehensive, and precise. Thus, no modifications were made, and they were presented in the same manner to the other participants.

The interviews, conducted with branch managers, deputy branch managers, supervisors, coordinators, members, and staff—each an integral part of their respective public institutions—lasted between 60 and 180 min and took place before the Kahramanmaraş earthquakes on February 6, 2023. In accordance with scientific ethical guidelines, participants' statements are included in the study without mentioning their names or affiliated public institutions, using only assigned codes. Table 2 presents the participant codes, ages, genders, the type of organization they belong to, and their positions within these organizations.

3.2. Data analysis

The data obtained in the study were analyzed through the MAXQDA 2020 program. The MAXQDA 2020 program is a platform used in the analysis of interviews and provides practicality in terms of time [96]. For this reason, the program was preferred and thus, the interview data were coded, themes were formed, analyzed, meaningful information was obtained, and the analysis was evaluated. In this context, firstly, the interview texts were coded by the researcher considering the research topic, questions, purpose and the

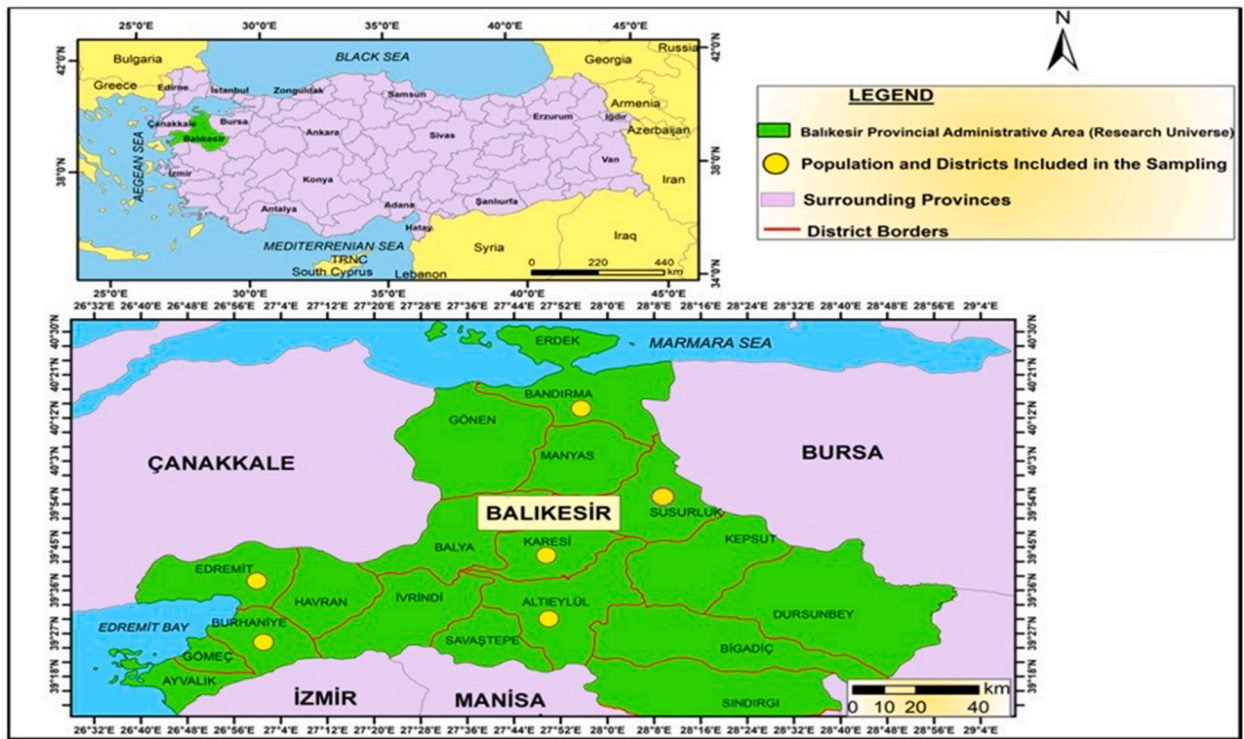


Fig. 1. Map of the research population and sample of public institutions.

Table 2
Participant codes and information

Participant codes	Age	Sex	Type of organization	Current status
P1	38	Male	Public Institution	Branch Manager
P2	46	Male	Public Institution	Acting Branch Manager
P3	44	Male	Public Institution	Supervisor
P4	27	Male	Public Institution	Member
P5	30	Male	Public Institution	Member
P6	48	Male	Public Institution	Coordinator
P7	50	Male	Public Institution	Expert
P8	56	Male	Public Institution	Branch Manager
P9	25	Male	Public Institution	Staff

theoretical framework of the research. As a result of the deficiencies seen in the first cycle coding, some sub-codes were removed, and new ones were added in the second cycle coding. As a result of the second cycle coding, a total of 108 sub-codes were created and these sub-codes were grouped under 18 codes. The 18 codes were grouped under 7 sub-themes and the sub-themes were grouped under a theme. Table 3 shows these themes, sub-themes and codes.¹

During the process of coding the interview transcripts, Saldaña's [97] work was utilized as a primary reference. This study highlights several critical aspects, including the philosophy of coding, the stages of coding in qualitative research, different types of coding, and their organization. Accordingly, these elements were taken into account in this research, enabling the implementation of a systematic coding process. In this regard, as previously mentioned, special attention was given to generating codes that were not only aligned with the research topic, objectives, questions, and conceptual framework but also engaging and non-ordinary. To ensure consistency, the codes, sub-themes, and theme were reviewed by two academics with expertise in the study's subject matter, and necessary revisions were made based on their feedback.

¹ Subcodes are not included in the table as they are too numerous.

Table 3

Codes, subthemes and themes used in data analysis

Theme	Subtheme	Codes
The Role of Civil Society Organizations in Disaster Governance from the Perspective of Public Institutions	Backgrounds of Public Institutions	Profiles of Public Institutions
	Roses and Thorns of Public Institutions	Public Institutions as a Panacea Public Institutions with Defective Production Public Institutions Speaking the Same Language
	Geographical Dimension of Disaster Organization	Integrity and Actions of Public Institutions in Balıkesir Geographical Context of Disaster Management
	Sustainability of NGOs and their Social and Internal Connections in the Eyes of Public Institutions	NGOs' Financial Assets and Survival Capacities NGOs Speaking the Same Language NGOs as an Antidote NGOs as a Poison Time Traveling NGOs NGOs Speaking Different Languages Social Aspects of NGOs
	Effects of Disasters on Disaster Organization and Disaster Perception	The Good Touch of Evil Standing or not Standing by NGOs Established and chronic disaster unawareness
	State and NGOs in Disaster Management Hierarchy	Determining the Relational Coordinates of the State and NGOs
	Seeing the Big Picture in Disaster Management	Taking a Picture of Disaster Management

4. Findings

4.1. Backgrounds of public institutions

4.1.1. Profiles of public institutions

An interesting picture emerges when the profiles of public institutions within the scope of disaster management in Balıkesir are evaluated. If their current names are considered, all institutions in the sample were established in the twentieth century. For example, Balıkesir Fire Department was established in 1926, while BBB Fire and Disaster Management was established in 2014. Another example is AFAD. Although the General Directorate of Civil Defense, which constitutes the foundation of AFAD, was established in 1958, AFAD was established in 2009. Therefore, the establishment years indicated in Table 4 are the dates of the current names of the relevant institutions. In this context, the oldest public institution is Balıkesir UMKE and the youngest is The Burhaniye Municipality Earthquake and Natural Disaster Commission.

When the number of members is analyzed, BBB Fire and Disaster Management has the highest number of members with 673 members, while The Edremit Municipality Disaster Affairs Commission has the lowest number of members with 3 members. AFAD Balıkesir, one of the most important bodies in disaster management, has 62 members (Table 4).

Considering the areas of expertise of public institutions, it is seen that they have different areas of expertise. It can even be mentioned that some institutions have many areas of specialization even within their own structure. In this context, the statement of a participant is as follows:

Researcher: Can you tell us about the specialization of your unit?

Table 4

Names, years of establishment and number of members of public institutions operating in the scope of disaster management in Balıkesir Province

Name of public institution	Year of establishment	Number of members
AFAD Balıkesir	2009	62
BBB Fire and Disaster Management	2014	673
Balıkesir State Hospital's Emergency and Disaster Unit	2016	180
The Occupational Health and Safety Unit of the Balıkesir Provincial Directorate of National Education	2015	55
The Civil Defense Unit of the Balıkesir Provincial Directorate of Agriculture and Forestry	2014	10
The Forest Fire Fighting Branch of the Balıkesir Regional Forestry Directorate	2012	23
The Burhaniye Municipality Earthquake and Natural Disaster Commission	2021	5
The Edremit Municipality Disaster Affairs Commission	2019	3
Balıkesir UMKE	2004	250

P9: You may have heard of crush syndrome. When a person is under the rubble for a long time, even if he/she comes out of the rubble with the effects such as crushing, injury, trauma, he/she dies 6–8 hours later due to conditions such as kidney failure. This was seen in the the Earthquake in 1999. With this professionalization in the health system and the emergence of our organization, life-saving interventions such as IV fluids are now being given to those trapped under the rubble. For example, this measure was very important to prevent crush syndrome. Apart from this, our organization also provides closeness with disaster victims. For example, there are experts in our team who provide psychological support. They establish intimate relationships with traumatized people. Our team includes people from many different professions such as specialist doctors, nurses, general practitioners, health officers, laborants, 112 personnel, midwives, underwater physicians (hyperbaric), home care technicians for the elderly, and drivers. All of these people are volunteers. We mostly work to improve public health in earthquakes.

Considering the multidimensional and complex nature of disasters, it is important that public institutions in Balıkesir province are diverse in terms of their areas of expertise and member qualifications. For this reason, the fact that almost none of the public institutions in the sample stands out in a single field creates the potential for increased efficiency in combating disasters. Because, as it can be understood from the examples of NVOAD and FEMA in the USA, the more diverse and inclusive the areas of expertise of the stakeholders in disaster management are, the more chances there are to increase the quality of the disaster organization [67].

Voluntary participation, one of the foundations of NGOs, plays an important role in many public institutions. In this sense, there are even public institutions that require volunteerism to take part in the teams affiliated to the unit. The 48-year-old representative added to his answer to the question on whether NGOs are given a voice in disaster management and stated the volunteering requirement of his unit as follows:

... All of our team are AFAD volunteers. In fact, we do not allow anyone who is not a AFAD volunteer to join the team. We have such a condition. We also have AFAD trainers in the team. (P6, Male).

On the other hand, there are also participants who emphasized the high number of volunteers affiliated to their units. At this point, one participant felt the need to answer questions about NGOs by associating them with volunteers in their own organizations and used the following statements:

Researcher: Are NGOs given a voice in state-based plans, projects or strategies related to disasters and disaster management?

P8: NGOs are not given a say in the plans. But they are definitely not excluded in their applications. For example, the president of the Young City Council applied to us to become a forest volunteer and received training. Apart from that, we have 1496 volunteers. In the Kepsut Karagöz fire in July 2022, there was a council of 10 people where volunteers worked. As I said, NGOs are not assigned tasks in planning, but their applications are accepted. (56, Male)

It is possible to say that public institutions do not take a position against civil society - at least not in its entirety - as they emphasize volunteerism by giving examples from themselves when asked about their perspectives towards NGOs. In addition to the presence of many volunteers in institutions with critical positions in disaster organization such as AFAD and BBB Fire and Disaster Management, as well as the experts working under public institutions in the sample, the fact that some participants stated the number of volunteers within their organization and frequently emphasized the importance of volunteerism provides significant data in this direction.

4.2. Geographical context of disaster management

Addressing disaster management both in Türkiye in general and Balıkesir in particular and making comparisons with other countries is another remarkable point of disaster management geography. Although the spatial characteristics of countries are different, there are approaches that should be adopted and taken as an example in disaster management. On the other hand, it is observed that only two of the representatives of public institutions made country-based comparisons. The observation made by the 27-year-old participant, who made one of these comparisons, includes a suggestion concerning disaster management as well as country comparisons. He responded to the question analyzing the problems in disaster management in Türkiye as follows:

... Apart from that, roofs should be removed from 5-storey buildings. You see in movies etc. in America, there is no such thing as a roof on buildings. Why should the roof be removed? Because tiles can fall in a storm. Damage to life or property. Pergolas (shades) can fall from terraces. These also pose a great danger. (P4, Male)

Different participants made a comparison between the USA and Türkiye and underlined the different practices in disaster management. Reminding that disasters other than natural disasters also affect Türkiye, the participant used the following expressions:

... As you may have researched, in the USA, for example, private insurance can come into play in the disaster law. There can also be different disaster laws in each state. It varies according to the disaster they are most affected by. For example, in some states, landslides are the focal point, in others, earthquakes or tornadoes. This is not the case with us right now. It is no longer in the dimension and focus of natural disasters. There is also humanitarian aid, and it is very important. Technological and humanitarian disasters, such as migration, are also very important. (P1, 38, Male)

Another subject emphasized by the participants in the geographical context is the aid teams whose locations vary according to the disaster area. Two public institution representatives expressed their opinions on this issue and emphasized the potential assistance that could come from neighboring provinces in the event of a possible disaster in Balıkesir. The statement of a participant on this issue is as follows:

Researcher: In your opinion, what kind of differences would there be in disaster management in the absence of NGOs?

P2: If the disaster occurs in Balıkesir, we as public institutions cannot do it ourselves anyway. Life comes first. That is why state institutions cannot provide for large-scale events. If it happens in Balıkesir, we will receive aid from neighboring provinces such as Manisa, Çanakkale, Bursa, İstanbul. But of course, it depends on the size of the disaster, or more precisely, the impact and damage caused by the disaster. (46, Male)

The geography of disaster management is perhaps one of the most important dimensions of the concept of disaster management. However, it is seen that only a few of the representatives of public institutions expressed their opinions on this issue. In one of these opinions, the existence of the motto “Near is my shirt but nearer is my skin” and in the other one, the evaluations that the assistance of NGOs in disaster management would not be much are interesting. On the other hand, in geographical terms, solidarity and activities of both them and other public institutions were mentioned more. In terms of the geographical aspect of disaster management, public institutions as well as NGOs have used the aid that can come from different regions according to the disaster area more than the comparison of the disaster management understanding of the countries. The fact that representatives from leading public institutions in Balıkesir’s disaster management emphasize aid from other regions in the event of a major disaster in Balıkesir aligns with the reality highlighted by Hermannson [17], namely that state-NGO collaboration in Türkiye is more prominent at the national level rather than the local level.

4.3. Sustainability of NGOs and their social and internal connections in the eyes of public institutions

Public institutions and NGOs, which are two different stakeholders in disaster management, have their own perspectives based on their perceptions arising from their relations with each other or from various factors. Naturally, these perspectives can be positive or negative. Since NGOs are at the center of the study and we want to understand what their roles are, determining the quality of NGOs in the eyes of a different stakeholder is more important and is explained most intensively in this section. In addition to the social dimension that constitutes the core of NGOs, sustainability and even the views of public institutions on the phenomenon of NGOs can be found in this section. Accordingly, the most prominent issue that public institutions emphasize about NGOs is *the toxicity* of NGOs.

4.3.1. The toxicity of NGOs

Public institution representatives think that there are many negative aspects regarding NGOs. In this sense, the most emphasized negative aspect is that NGOs put themselves at the center of the disaster. Many of the participants mentioned that NGOs have the disease of putting themselves at the forefront in disaster activities. For example, a 56-year-old representative, in response to the question of what kind of capacity differences exist between public institutions and NGOs in disaster management, mentioned the tendency of NGOs to present themselves differently from what they are and said the following:

I think NGOs have a tendency to exaggerate their activities. The main work is done by state institutions. But NGOs try to stand out. We state institutions do many times more than what they do. (P8, Male)

Other 44-year-old participant (P3) talked about the irresponsible behavior of NGOs. In response to the question of what the advantages and disadvantages of NGOs in disaster management are, the participant mentioned the existence of NGOs that prioritized making their own advertisements after the earthquake in İzmir on October 30, 2020 and those who tried to take “selfies” in the rubble.

Four public institution representatives directly criticized AKUT by name and claimed that AKUT was trying to gain publicity. One of the participants also stated that AKUT’s negative attitude changed AFAD’s perspective on NGOs. The evaluation of a participant on this issue is as follows:

Researcher: Are NGOs also given a say in state-based plans/projects/strategies related to disasters and disaster management?

P7: ... But NGOs should not act competitively and should be a complementary element. For example, AKUT contributes 1 percent, but they try to stand out. They stand out. (50, Male)

Similarly, a different participant, while referring to the missing NGOs, argued that the free structure in NGOs brought along some problems and that AFAD accreditation emerged because of these problems. The comments of the 38-year-old participant, who added to the question of how their relations with NGOs are, are as follows:

There is also freedom in NGOs. Because of this freedom, sometimes negative situations could arise. For example, in the İzmir Earthquake, 150 people were found in the rubble. In this case, AFAD was completely to blame in the eyes of society. There was no negative criticism against NGOs. To prevent such situations, AFAD now has an accreditation system and those who can provide this accreditation can participate in disaster processes. (P1, Male)

Another issue that stands out regarding the negative aspects of NGOs is the tendency of NGOs to act in an irregular manner unless they are led by a leader. This tendency also means that NGOs will not be able to carry out disaster management efficiently if the state is insufficient. One participant’s answer to the question whether NGOs can provide the necessary efficiency in disaster management in case the state is insufficient is as follows:

It is very difficult for them. They cannot cope with events in disasters. It would be like Armageddon. Besides, NGOs have fewer numbers and personnel. Therefore, I think it cannot happen. (P7, 50, Male)

Another crucial negative aspect of NGOs is the insufficient vision of NGOs. According to the participants, NGOs' insufficient vision leads to wrong attitudes on various issues. One of the most striking observations about insufficient vision is that the difficulties in communicating with some NGOs stem from the low quality of the NGOs in question. A 38-year-old participant mentioned this issue in response to the question whether there are obstacles limiting the contribution of NGOs to disaster management and made the following assessments:

Our contact with NGOs can be limited. But let me tell you how. Our communication with NGOs that will not be competent and efficient in participating in disasters is limited. Sometimes, for example, there are interesting situations. It has been 13 years since we opened the NGO, but when we call the president of this NGO, he says, "Did we have such an association?" (P1, Male)

Another issue that stands out as one of the negative aspects of NGOs is NGOs that engage in disinformation. There were two participants who underlined this issue. The answer given by one of the participants to the question of 'what are the disadvantages of NGOs in disaster management' is as follows:

As a disadvantage, prejudiced people within NGOs can exaggerate any flaw in the aid provided by the state to the public on social media. Or even if there is no defect, they can reflect a normal practice or procedure as not extinguishing the fire. (P8, 56, Male)

It is evident that participants attribute many negative aspects to NGOs. Especially, public institutions are seen to be more altruistic than NGOs. At this point, P8's statement that public institutions contribute much more to disaster management than NGOs and P2's emphasis on the difficulties of obtaining permission are striking approaches. Moreover, in the eyes of most participants, AKUT is the embodiment of the negativity of NGOs. The high number of participants targeting AKUT and citing it as an example of negative NGO attitudes is striking. P2, criticizing AKUT, claimed that AKUT misled the public and engaged in disinformation, while P5, P6, and P7 argued that despite AKUT's insufficient contribution, it attempted to highlight itself and promote its own image. Apart from all these, it is also thought-provoking that the insufficient vision of NGOs in the eyes of public institutions reveals different wrong behavior models. In particular, the example of a president who is unaware of the existence of his association, mentioned by one of the participants, is a contribution to the previously mentioned determination of signboard associations and reveals that the poison of signboard associations should be removed from the bodies of NGOs.

4.3.2. The antidotes for NGOs

The antidotes of NGOs have been mentioned considerably, although not as much as their poisons. At this point, the most prominent issue is NGOs' assistance to public institutions. Participants mostly associated NGO assistance with the provision of extra labor (Table 5). In response to the question of whether NGOs have a role to play in turning the crisis environment created by disasters into an opportunity, one of the participants gave an answer linking NGO assistance with public trust and made the following assessments:

Not to turn it into an opportunity, but the more officials in the disaster area, the better. Because as the number of officials increases, people feel safer. For example, when there is 1 official per 100 square meters, they feel afraid. But when there are 10 officials per 100 square meters, they feel safe. In this respect, NGOs are important for personnel support. In addition, NGOs do whatever task they are given. When we ask them to distribute soup, they do that too. (P9, 25, Male)

Another participant emphasized the spatial dimension of NGO assistance to public institutions. The comments of the participant in response to the question analyzing the advantages of NGOs in disaster management are as follows:

NGOs are local elements. People in a region establish NGOs. They naturally know where they live. In this sense, they guide public officials or other NGOs that reach there in the event of a disaster. They reflect values. It is very difficult for state personnel coming from other places to know the place as well as an NGO member. NGOs also provide aid in places where the state cannot reach. (P7, 50, Male)

The labor contribution of NGOs to disaster management is another important issue emphasized by the participants. A 38-year-old participant's expression of the contributions of NGOs in various fields is an interesting example in this sense. In response to the question investigating the roles of NGOs in disaster management stages, the participant expressed the following opinion:

Different professional groups in NGOs are also our richness. The longest and most tiring phase of the disaster process is recovery. They contribute to us in this respect. Apart from that, they even help us with document registration, which is very important. NGOs also provide nutrition aid and in-kind donations. We always want this to continue.

Table 5

Numerical code relations scanner showing the interrelationships between the subcodes of NGOs' assistance to public institutions and NGOs' provision of extra labor.²

Code system	NGOs' assistance to public institutions	NGOs providing extra labor force
NGOs' assistance to public institutions		14
NGOs providing extra labor force	14	

² The number reflected here shows the extent to which the relevant codes were used together in different words, sentences or sentence groups in the interview texts based on the interviews with public institutions.

It is mostly the older members of NGOs who want to participate in the wreckage work. Young people take a more holistic view. They want to participate in processes such as food and clothing distribution. They want to take part in important tasks such as accounting records. For example, recently an NGO member came to our branch and distributed *Magnum* ice cream to all the staff. It seems small, but these are important things. (P1, Male)

In parallel with the negative aspects of NGOs, there are also very different and diverse aspects in their positive aspects. In other words, in the eyes of public institutions, the contributions of NGOs to disaster management are not to be underestimated. Indeed, the fact that even the public institution representatives who emphasize the poisons of NGOs the most do not reject NGO contributions and even state that NGOs are needed in some cases is one of the most important indicators of this. Apart from all these, the fact that a participant emphasized NGO members who have the skills to provide local guidance to both public officials and other NGOs is a noteworthy observation in the geographical context. Because, as explained in the literature, the local knowledge of NGOs is a quality that creates added value for all stages of disaster management [94,98,99].

4.3.3. NGOs in terms of their financial assets and survival capacity

In addition to NGOs having a sustainable structure, their full-performance contribution to disaster management depends on many variables. However, the most important of these variables is the economy. Representatives of public institutions also have a perspective in this direction. Because the most important issue that participants put forward regarding the survival of NGOs is the fight against inadequacy. According to the participants, NGOs' fight against inadequacy essentially means an economic war. In this regard, one participant argued that NGOs operate with their own means and used the following sentences:

Researcher: Do you think there are barriers that limit the contribution of NGOs to disaster management? If there are, what are these barriers?

P9: I think there is an economic barrier. I think that efforts should be made to integrate NGOs, not politically. We need to act together.

Researcher: Can you give examples of economic barriers?

P9: There is a lack of funding. NGOs cannot receive funding for disaster activities. For roads, food, etc. They have to provide this with their own means. (25, Male)

Some participants associated the economic inadequacies of NGOs with the necessity to use imported materials. In this context, one participant's views are as follows:

In addition, the materials that NGOs need to use in disaster activities are also foreign-sourced and expensive. For example, a meter of rope costs I don't know how many euros, but you can only use it three times. Then throw it away. Even small pieces are expensive. (P6, 48, Male).

The common view of public institutions is that NGOs have extremely limited resources. The high level of awareness of public institution representatives regarding the economic barriers that NGOs face is also striking. The fact that two participants emphasized the issue of lack of funds is particularly important in this respect. Because one of the main conditions for producing solutions to problems is to identify the source or sources of the problem. Therefore, it is important for the participants to sincerely express problems that are not directly under the responsibility of NGOs and are of state origin in terms of the existence of solution-oriented thinking.

4.3.4. Social aspects of NGOs

The fact that their ultimate goal is to benefit society, which is also their foundation, makes the social aspects of NGOs extremely

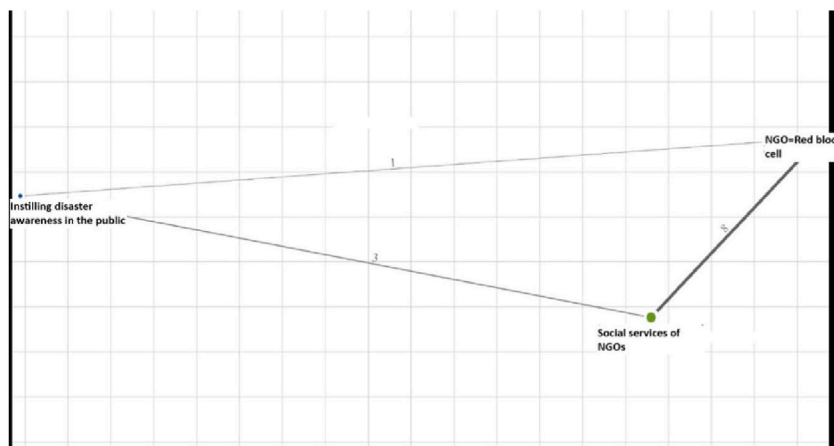


Fig. 2. Code map showing the relationships between *instilling disaster awareness in the public*, *social services of NGOs* and *NGO = Red Blood Cell* subcodes.

important. Moreover, this issue is one of the issues that many participants have underlined. Moreover, it is understood that the issues of NGOs raising disaster awareness in the public, NGOs' social services and red blood cell (social improvement) features are closely related to each other (Fig. 2). In the context of NGOs' social services, it is seen that the answers of two participants are in response to the question of whether NGOs have an impact on post-disaster development processes. In this context, one participant emphasized the rehabilitation aspects of NGOs, while the other participant used the following expressions:

NGOs have an impact on development. For example, a school may be destroyed in an earthquake. NGOs like Eğitim-Sen (*Education Syndicate*) set up tents and provide education there. They do this while the state is building a new school building. (P4, 27, Male)

The red blood cell characteristics of NGOs are another issue adopted by the participants. Another participant, aged 48, points out the broad scope of the role of NGOs in the red blood cell aspect. In response to the question of what the duties of NGOs are in the post-disaster development period, the participant's views are as follows:

NGOs have a role in overcoming social traumas after disasters. They provide psychological support. They also help in the establishment of tent cities, such as food distribution. In terms of economic development, NGOs can open vocational courses in disaster areas, for example. They can make such a contribution. They can contribute to social development by providing rehabilitation. For example, agriculture, which is the most important source of income in a disaster area, has died after the disaster. NGOs can help society find other sources of income. (P6, Male)

The participants' evaluations reveal that the social aspects of NGOs are extremely important both before and after the disaster. The thesis in question is also parallel to the perspective of Benson, Twigg and Myers [94] and Shaw [100]. However, based on the evaluations of a large portion of the participants, the following addition should be made: Despite the seemingly indispensable roles of NGOs in Balıkesir province in disaster management stages, the extent to which they are successful in implementing these is debatable in the eyes of the participants.

4.3.5. NGOs speaking different languages

NGOs have different aspects. It is natural to compare these differences, which include many variables. In fact, the participants felt the need to compare the distinctive features of NGOs and expressed their opinions accordingly. The most frequently mentioned topic was that the levels of success vary according to NGOs. A 44-year-old participant made an evaluation of NGOs throughout Türkiye rather than specific to Balıkesir province and adopted an approach that highlighted AKUT and presented AFAD and AKUT as equivalent. In response to the question of how the position of NGOs in disaster management has changed in the last 20–25 years, the points emphasized by the participant are as follows:

Of course, AKUT is very high-level. In fact, I think AKUT and AFAD are equal. AFAD wants to work with institutions such as AKUT and Gendarmerie Commando Search and Rescue (JAK). In fact, there is a personnel flow between AFAD and AKUT. Those who retire from AFAD transfer to AKUT. Those who leave AKUT transfer to AFAD. AKUT is also faster than other NGOs in terms of time. Therefore, AKUT is the NGO that stands out. (P3, Male)

There are also participants who compare the success levels of NGOs specifically in Balıkesir province. In response to the question of how their relations with NGOs are, a participant made the following comments:

... But for example, we are pleased with Gönen Natural Disasters Search and Rescue and First Aid Association (GÖNDAK) in Balıkesir. They work with discipline. So are The 911 Search and Rescue Association and Dursunbey Search and Rescue Association (DAK). (P2, 46, Male)

Although some of the participants heavily criticize AKUT, as understood from this section, it is also possible to talk about those who see AKUT as more successful and separate it from other NGOs. Therefore, in addition to not reaching a minimum consensus regarding AKUT, there is either complete criticism or complete praise. It is also striking that members of the two most vital institutions of disaster management stated that a few of the disaster-themed NGOs in Balıkesir province are valuable and useful. It is likely that this situation is related to the abundance of signboard associations in Balıkesir province in the eyes of the participants.

4.3.6. NGOs traveling in time

How the roles of NGOs in disaster management have changed, especially in the last 20–30 years, is a subject commented on by some participants. Participants who made evaluations on the relevant question prioritized the changes in the quality and quantity of NGOs. A 50-year-old participant expressed the development of NGOs in terms of quantity and quality and the source of this development as follows:

Of course, there were very few NGOs 20–30 years ago. With the development of democracy, technology, social media, etc., NGOs have also increased. People connect to NGOs with a sense of belonging. They feel safe here. In addition, NGOs have developed both in number and in terms of workload, that is, personnel. (P7, Male)

Another participant emphasized the 1999 Gölcük Earthquake and argued that NGOs have developed in terms of quality in different disaster processes and areas. The participant's statements are as follows:

The 1999 Earthquake is a turning point in this respect. It is also a turning point for the world. There was search and rescue, but medical rescue spread all over the world after this earthquake. NGOs were only providing benefits during the recovery period. Now they have developed a lot in intervention, first aid, search and rescue and they are providing benefits. (P9, 25, Male)

A 48-year-old participant brought a different dimension to the subject. While highlighting the development of NGOs in terms of number and quality, the participant also said that NGO members have made civil society culture a lifestyle. The participant's sentences are as follows:

The participation of young people and university students in NGOs has increased. Thus, both the number of NGO members and NGOs has increased. The literacy rate used to be low. Now it is high. People continue NGO activities and what they have learned in their private lives, politically and culturally. (P6, Male)

It is understood that the views expressed on the journey of NGOs over time are mostly positive. Because most of the participants who touched on the subject argue that NGOs are developing in terms of quantity and quality. In addition, a participant's evaluation of the numerical increase of NGOs because of democracy and technological development refers to Castells' [101] concepts of information society and information revolution.

4.3.7. NGOs speaking the same language

Contrary to the different aspects of NGOs, their communication and mutual assistance were only mentioned twice by two participants. The views that can be evaluated within this scope consist of some statements that were also shown in the previous sections. While the 50-year-old participant drew attention to the geographical guidance between NGOs, the 46-year-old participant evaluated the issue as follows:

Researcher: Do you think NGOs have an impact on the development processes after a disaster? If so, what are these impacts?

P2: They can help not only after a disaster but also before a disaster. They help their members gain disaster awareness. (46, Male)

According to the participants' perspective, it is seen that NGOs speaking the same language is shaped by two elements. These are communication and information transfer between NGOs. Naturally, these factors not only develop in relation to each other, but also increase the quality of NGOs. However, it is understood that the important issues in question are both expressed by a small number of participants and their statements remain relatively shallow. Therefore, it is possible to talk about some deficiencies and problems in the perspectives of those who do not mention the subject, as well as those who do.

4.4. Determination of the position of the state and NGOs in the disaster management hierarchy

As mentioned in previous sections, one of the most critical factors determining the position of NGOs in disaster management is -perhaps the most important-their relationships with public institutions. At the same time, these relationships provide significant data on public institutions' perceptions and attitudes towards NGOs. Therefore, in the eyes of public institutions, the evaluation of their contacts with NGOs and their positions in decision-making processes is at least as necessary as determining the NGOs' perspective on the issue. In this sense, the issue that public institutions emphasize the most is that the state is stronger than all other stakeholders in disaster management.

4.4.1. Relational coordinates of the state and NGOs

It is observed that the participants emphasize the power of the state, which is incomparable to NGOs, in the context of the state-NGO connection. Since it is the reflection of the state in disaster management, AFAD also represents the embodiment of this power. For this reason, the fact that the state has the strongest position in disaster management also means AFAD's superiority over NGOs in the hierarchy and AFAD's authority to take the initiative in participating in disaster management. The following thoughts of the 38-year-old participant are an example that sheds light on these three points. The participant, who added in his answer to the question of whether there are obstacles that limit the contributions of NGOs in disaster management, defended the following views:

We always get a response when we ask for support. As I said, NGOs are the cogs of the machine. Disaster management should be under the monopoly of AFAD. NGOs should only implement the given directives. The disaster emergency management center is the kitchen of this business. Everything is here. Logistics materials, communication etc. (P1, Male)

A 25-year-old participant also expressed his views on the superiority of AFAD. The participant responded to the question of whether NGOs were given a say in disaster management as follows:

I think they were given a say. But it is important for NGOs to act according to the instructions given by AFAD. I think we need to think from this perspective. (P9, Male)

Another participant generalized AFAD's hierarchical superiority and initiative authority to all public institutions. The participant's thoughts on whether NGOs created a disadvantage in disaster management were as follows:

Their willful and undisciplined actions. For example, we sometimes tell them *not to leave this area and intervene here*. But then we see that they have gone somewhere else and are working there. Such situations create negativity. NGOs need to be under the management of public institutions. (P2, 46, Male)

The same participant also made an original assessment to emphasize the power of the state against NGOs. In response to the question of whether there are economic or political obstacles that limit the contribution of NGOs to disaster management, the participant made the following comment:

There is no political barrier. I think the materials and equipment that NGOs need should be found in state institutions. In this respect, NGOs should learn to use these materials. They should work under public institutions. I don't know if this is the right expression, but they should obey. (P2, 46, Male)

One of the most frequently mentioned points by the participants is that public institutions should be in dialogue with NGOs. A 38-year-old participant gave an interesting answer to the question of how their relations with NGOs are. The participant's comments are as follows:

We have good communication with NGOs. We are like siblings in a way. (P1, 38, Male)

Another 27-year-old participant emphasized the positive relationship they have established with an NGO in their region. In response to the question of what kind of work they have done regarding disasters since their unit was established, the participant shared his thoughts as follows:

... We conduct disaster location studies with amateur radio operators. In order to understand the areas that are damaged in a disaster. We are communicating with the amateur radio association in Edremit regarding this issue. (P4, Male)

Some participants believe that public institutions are much more devoted than NGOs. For example, a 46-year-old participant emphasized the professionalism of public institutions and the amateurism of NGOs. The participant who answered the question of whether there are obstacles that NGOs face in disaster management by adding the following assessments:

... There is no luxury of being carried away by state institutions. For example, we often cannot even take leave in the summer. We cannot go on vacation. We return from leave even when we do. There is no such situation in NGOs. They are volunteers after all. They have other jobs. They may have the luxury of not participating in disaster activities because of their jobs. (P2, Male)

Another participant who highlighted the dedication of public institutions argued that, unlike NGOs, the principle of "*Near is my shirt but nearer is my skin*" applies to them. The participant used the following expressions:

People working in NGOs are family men or members. They put their children or spouses first. They see disasters as secondary. People working in the state do not have such a situation. Even if they have families, they see disasters first. In NGOs, life comes first, that is, their own lives and the lives of their families. Then comes the beloved. In the state, it is the opposite. (P7, 50, Male)

The different working principles of public institutions and NGOs were another emphasized point. A 38-year-old participant compared the structural differences of NGOs with their own. When asked how the position of NGOs in disaster management has changed from the recent past to the present, the participant added the following points:

As our institution, we can neither keep people close nor far away. According to their bylaws, NGOs are only oriented towards helping Turks in disasters. However, according to the law, we are obliged to help everyone, regardless of whether they are Turks, Syrians, Greeks, young, old or children. We even sent a team to the nuclear explosion in Japan. There was no one from the so-called superpowers, the USA or Russia, in this disaster, but we were there. In this respect, our institution's presence is very valuable. (P1, Male)

The presence of four participants who think that NGOs are not given a say in disaster management is striking. Perhaps the most important view at this point was shared by the 27-year-old participant. The participant commented on the question of whether NGOs are given a say in disaster management as follows:

No, NGOs are not given a say. Because NGO representatives do not have legal guarantees. Even if an NGO representative uses the wrong address, they can be penalized. An NGO representative who knows this cannot say anything to the central structure. People naturally think of themselves and their families before anyone else in such situations. (P4, Male)

It is not a coincidence that the power of the state in disaster management is the most frequently mentioned issue by the participants. Because by mentioning this issue, the participants wanted to reveal the hierarchical superiority of public institutions over NGOs, that the state, not NGOs, directs disaster activities, and that the state has all the means. The reasoning behind the desire to state these points is as follows: "*There is a limit to what NGOs can do in disaster management. Therefore, NGOs are obliged to act both by accepting this fact and with the awareness that there is a state stronger than them. Because although NGOs are needed in disaster management, the real perpetrator and guiding party is the state.*" A similar approach can also be observed in countries such as Nepal, Uganda, and Pakistan. In these countries, governments consider the inclusion of civil society's feedback and opinions in disaster strategies to be nearly impossible. As a result, they prioritize public institutions in decision-making processes, keeping them in a dominant and powerful position [16,102]. This understanding is entirely contrary to the bottom-up disaster management and disaster (risk) governance models, where all societal stakeholders are considered equal to the state in decision-making processes. Furthermore, as Dipendra [64] emphasizes, it reveals that

the concept of governance in developing countries has not matured as much as in developed countries and, naturally, that the 'paternalistic state' mentality in Türkiye has also influenced Balıkesir.

4.5. The impact of disasters on disaster organization and disaster perception

Many disasters have occurred in Türkiye throughout history, and it is possible to talk about the potential for future disasters. Therefore, Türkiye is in a sensitive position against disasters. Due to this situation, individual, social, economic and political perspectives on disasters and the concept of disaster are constantly changing and transforming. In this context, representatives of public institutions have mostly focused on whether or not NGOs are supported and the impact of disasters on the relevant issue.

4.5.1. Support or not to support NGOs

It is seen that many evaluations are made regarding the support given or not given to NGOs. In these evaluations, both the social and political dimensions of the support are addressed. In this respect, the participants mostly emphasized the support of NGOs by public institutions. The 38-year-old participant revealed the importance of the integrated disaster management system in terms of the support they provide to NGOs. The participant answered the question of whether NGOs are given a say in disaster management as follows:

Within the scope of the integrated disaster management system, we mostly cooperate with citizens and NGOs. Of course, they are given a say in this sense. But NGOs are part of the machine. They have to implement AFAD's directives. (P1, Male)

Another 44-year-old participant emphasized the support AFAD provides from time to time to compensate for the deficiencies of NGOs struggling with impossibilities. In response to the question of whether there are obstacles that limit the contribution of NGOs to disaster management, the participant shared his thoughts as follows:

They have economic obstacles. They do not have enough technical equipment for search and rescue, first aid, etc. This equipment is also very expensive. For example, the rescue vehicle costs around 1 million liras. However, AFAD allows NGOs to use its own equipment when necessary. I do not know how much funding the state gives them. (P3, Male)

Another frequently mentioned issue is that the public does not know NGOs well enough and therefore does not trust them. In this regard, the 56-year-old participant's emphasis on the Red Crescent and the Human Rights and Freedoms Humanitarian Relief Foundation (IHH) is striking. In response to a question examining the trust level of the local public regarding the capacity and role of NGOs in disaster management, the participant's evaluations are as follows:

I don't think they know enough. There is also a trust problem. That's why they don't feel respected. I won't say it now, but associations like the Red Crescent and IHH have had some problems in the past. I think that's why people don't feel trusted. (P8, Male)

In response to the same question, another participant highlighted the problems in the education curriculum and the political structure. The participant's statements are as follows:

For example, ask the people in this café, no one knows about NGOs. Because there is an education system that prevents NGOs from getting organized. They don't teach what civil society is, the necessity of civil society, solidarity, responsibility, etc. People can't even hand out brochures because they are afraid of the political environment. (P4, 27, Male)

The support of public institutions to NGOs was mentioned significantly more than the thesis that they did not support. The main actor in this support was AFAD. Because a significant number of participants argued that state-based aid to NGOs was mostly provided through AFAD. However, when it comes to the public's perspective on NGOs, there are problems that need to be underlined. As some participants pointed out, the fact that the Red Crescent, in particular, has a bad criminal record in the eyes of a significant number of the public has led to a negative perception of all NGOs. It should be admitted that when such a perception exists, it is very difficult for the situation to suddenly reverse and for people to trust NGOs. Çakı [21] highlighted the issue of distrust toward NGOs in Balıkesir in his study. The author also identified one of the reasons behind the negative perception of NGOs as their associations with organizations like the Fethullahist Terrorist Organization (FETÖ), which orchestrated the attempted coup on July 15, 2016, and had connections with NGOs operating in various fields, including disaster response. As a result, not only the political mechanisms but also a significant portion of society became wary of NGOs. When this study is considered alongside the findings of the present research, it becomes evident that there is a problematic environment in terms of public trust and support for NGOs.

4.5.2. Disaster unawareness becoming established and chronic

One of the most critical thresholds in combating disasters is the high level of disaster awareness in society. However, participants mostly complained about the problem of lack of disaster awareness in society. The 50-year-old participant brought disaster unawareness to the forefront in terms of ensuring people's awareness of helping each other in vital processes after a disaster. The participant added the following statements to his answer to the question of whether there is a capacity difference between public institutions and NGOs in disaster management:

... When there is a major disaster, all state institutions cannot go to the disaster place at the same time and within 72 hours. People need to be self-sufficient during these golden hours. This disaster aid awareness needs to be created. (P7, Male)

The 30-year-old participant brought a different perspective to the subject by evaluating the lack of social disaster awareness in terms of disaster gathering areas and theft. The participant's answer to the question of whether there is a problem in disaster management is as follows:

... You may have heard that there was a flood in Edremit in 2013. We were stranded in front of Kipa Market in the flood. People didn't know where the safe places were back then either. Nothing has changed today. For example, traffic signs or road signs can be put up for disaster assembly areas. But even the disaster assembly area signs here are being stolen. There is no proper plan or program for this reason. (P5, Male)

Another 27-year-old participant defended the same thesis with different sentences. The participant answered the question examining the activities of their units as follows:

We worked to implement penal sanctions for column cuts in buildings with shops on the ground floor. In addition, our people do not know about disaster assembly areas. As you know, there are cases of theft of signs. Therefore, we worked to ensure that our people knew about disaster assembly areas. We had maps of disaster assembly areas hung at the mukhtar's offices. (P4, Male)

Another issue that the participants highlighted was the transience of the importance given to NGOs. According to the 44-year-old participant, this problem also means that disasters are not given any importance or that the importance given is temporary. The participant answered the question examining the public's trust in the capacity of NGOs in disaster management as follows:

Well, you know our people. An incident or disaster needs to happen for awareness to be created. Disaster awareness is created like this. Only after a disaster. I think the local people do not know about NGOs and do not trust them. Their trust is in AFAD and health institutions. They already know about these institutions when an incident occurs. (P3, Male)

To emphasize the lack of disaster awareness, the participants mostly underlined the looting and theft incidents that occur after disasters. In fact, it can be seen that incidents such as the theft of directional signs can occur even before a disaster. In fact, it is difficult to explain this situation only with a lack of disaster awareness, and it also has humanitarian and moral aspects. Therefore, the fact that even a minority of society has deficiencies in these areas creates a potential problem in combating disasters.

4.5.3. *Evil reveals good*

Some of the participants, unlike the determination of lack of disaster awareness, argued that the perspective on disasters has changed both socially and politically and disaster awareness has increased as lessons are learned from large-scale disasters. In other words, the participants in question are of the opinion that it is possible to evaluate the crisis environment created by large-scale disasters as an opportunity. For example, the 56-year-old participant mentioned the change in mentality in the state as a result of AKUT coming to the fore after the Kocaeli-Gölcük earthquake in 1999. In response to the question of how the position of NGOs in disaster management has changed in the last 20–30 years, the participant's views are as follows:

I think NGOs are successful especially in the prevention and awareness-raising stages of disaster management. They are partially good in the combat stage. For example, AKUT has dogs that it uses in operations. They provide benefits. The state has thus pulled itself together. Before the 1999 Earthquake, there was only the Civil Defense Unit, and in my opinion, it was worthless. It had no use. But after it became AFAD, its power to combat disasters increased. (P8, Male)

The 38-year-old participant made important observations while revealing the change in mentality in Türkiye regarding disasters. The participant made the following statements while providing information about the institution he is affiliated with:

Our institution is focused on disasters and emergencies. We attach great importance to pre-disaster preparation. In previous periods, we, as a society, were only focused on the post-disaster period. But pre-disaster preparation is very important. We have an "Integrated Disaster Management System" model for this. This model aims to identify hazards and risks, prevent or minimize possible disaster damage, ensure effective intervention, and also provide post-disaster recovery efforts that we attach great importance to in an integrated manner. In this context, we also provide training 24/7. The pandemic did not affect this process either. We provided online training via Zoom. (P1, Male).

When the participants' views are evaluated, it can be said that especially the Gölcük Earthquake in 1999 and the İzmir Earthquake in 2020 were turning points in the change of perspective on the concept of disaster and disaster organization. For example, within the scope of disaster organization, the emergence of compelling conditions for the establishment of AFAD after the 1999 Gölcük Earthquake and the emergence of the AFAD accreditation system following the 2020 İzmir Earthquake have been radical changes for disaster organization in Türkiye. It is also understood that social and political changes in the perception of disasters and disaster perception mutually affect each other. In particular, the emphasis of a representative of an institution with an important place in disaster management, which emphasized that while there was a tendency in the past to focus on the post-disaster period in society, the importance of the pre-disaster period was later understood, is striking at this point. Indeed, this change in mentality has also affected the pre-disaster activities of public institutions and has revealed a noticeable increase in these activities. The increase in relevant activities naturally means that public institutions convey the importance of the pre-disaster process to more people.

4.6. *Seeing the big picture in disaster management*

Just as early diagnosis is extremely important in the treatment of fatal diseases, it is equally important to identify problems (if any) in the disaster organization and take steps to solve them before disasters occur. A holistic perspective is essential for identifying and

solving problems. In other words, it is essential to see the whole picture. What public institutions, who are decision-makers in disaster organizations, think about the organizational framework will make it possible to determine their ability to see the big picture. In this sense, it is noteworthy that the most emphasized issue by the participants is the strategies that need to be developed for critical elements in terms of disaster.

4.6.1. Pressing the shutter button to determine disaster management

Disaster management has been presented with its pros and cons by the participants. Creating strategies for disasters and issues that are believed to need to be addressed urgently for disaster management is a subject that the participants frequently express. It is possible to talk about the relationship between the strategy production thesis and the legal dimensions of the weaknesses in disaster management. The 56-year-old participant questioned the contributions of NGOs as part of disaster management and established relations with the work of public institutions. The participant's addition and evaluation regarding the question of how the absence of NGOs affects disaster management is as follows:

... For example, in the studies carried out by the state to reduce the response time to a disaster from 10 minutes to 8 minutes, NGOs should contribute to studies such as R&D. In other words, NGOs should serve to raise the standard. They should offer new solution proposals. For example, we are working on this issue as a state. We are discussing how to reduce the time from 15 minutes to 10 minutes in the event of a disaster in Balıkesir. (P8, Male)

Another participant also drew attention to the academic deficiencies in Türkiye regarding disasters. The participant defended the importance of specialization in this regard. The participant shared his thoughts as follows:

Researcher: Do you think there are problems in disaster management in Türkiye? If there are, what is your solution proposal?

P4: Of course, people would not send us IBANs if there were no problems. For example, is there an educational institution related to natural disasters? Is there a university? Are expert academics trained for this department? Academics determine the fate of a country. Even politicians train academics. They are of great importance. Since there are no such institutions, there may be problems in terms of merit. Accountants, economists, etc. will be appointed to head AFAD and the Red Crescent. For example, is it possible for an accountant or economist to know which forests could burn in a heat wave? But only people who are knowledgeable about natural disasters can do this.

There is another participant who draws attention to a problem originating from NGOs in disaster management. In addition to the question of whether the public trusts NGOs regarding their capacity and role in disaster management, the participant used the following sentences:

... NGOs need to be disciplined and directable. They need to be under the guidance of public institutions. I think the activities of NGOs should be controlled by law. (P2, 46, Male)

Some participants drew attention to the effect of unhealthy construction in disasters. In this context, two public institution representatives expressed the inadequacy of urban transformation. The views of one of the participants are as follows:

Researcher: Do you think there are problems in disaster management in Türkiye? If so, what is your solution proposal?

P9: Of course, even if we are prepared for disasters or say so, the impact of disasters is very important. The impact of disaster determines everything. After all, one can never be 100 percent prepared for disasters. Urban transformation is not sufficient in Türkiye now. It is at most 3 %, 5 %. It is 10 % at the most. This percentage needs to be increased. (25, Male)

The evaluations of public institution representatives show that - except for one person - there is a common view that there are some problems regarding disaster management in Türkiye. Although the NGO-based dimensions of the problems in question are mentioned, it can be stated that their political and legal basis are more prominent. In this context, problems such as deficiencies in urban transformation or the lack of systematic training provided in terms of increasing disaster awareness were mentioned. However, it cannot be said that the elements taken into the frame while taking photographs of disaster management are completely negative. Because two participants shared their thoughts that disaster awareness is increasing in Balıkesir.

5. Conclusions and discussions

Disasters carry risks that can threaten the existence of individuals and communities as well as societies and therefore countries. In fact, a disaster process that is not managed well has the potential to cause chaos that will spread throughout society. One of the most important conditions for good disaster management and success in combating disasters is a disaster approach that encompasses the social base. The approach in question is a bottom-up disaster "management/governance" model that embraces pluralism, different views and stronger democracy, rather than a top-down disaster management dictated by the ruling power to the people [44]. Because the complexity of disasters in terms of both their causes and consequences necessitates an all-out struggle. For this reason, the characteristics of disaster management systems in effect in countries are directly related to how democracy, which is the most suitable model for all-out participation, is viewed and internalized. One of the most important assets of democracy is civil society. Because civil society, true to its name, claims to represent all individuals in society and is in line with the main principles of democracy. However, there is a "Paternal State" tradition that has been valid in Türkiye for centuries. It is inevitable that disaster management, which is a part of state policies and greatly concerns society, will also receive its share from the ongoing "Paternal State" understanding. In

addition, as argued in the social constructivist disaster and risk paradigm, one of the most important determinants of disaster risk is the power relations that exist and are reinforced in society. These power relations are also shaped by the “*Paternal State*” tradition and cause civil society to remain in the background. Therefore, the state of civil society reflects democratic deficiencies and points to the dominance of a top-down disaster management that increases disaster risk. Although it is accepted that NGOs provide labor in public institutions, the view that the absence of NGOs will not disrupt the functioning of disaster management is widespread. In other words, it is thought that in the absence of NGO contribution, the process will be prolonged, especially in search and rescue and all other aid after disasters, but the state will ultimately be successful. However, it can be said that the situation did not work this way in the Kahramanmaraş earthquakes that occurred on February 6, 2023. The inadequacies that emerged after the earthquakes made international aid inevitable. International NGOs also played a significant role in the aid in question. In addition, at this point, the opinion of the AFAD representative, the highest body in disaster management, differs from the dominant perspective of public institutions. The AFAD representative believes that even the smallest contribution of NGOs in disaster management is extremely important. Therefore, it can be said that despite the dominant view of public institutions, the problems in disaster management will increase without NGOs. AFAD’s perspective in question is also parallel to the literature. Indeed, the vital and difficult to replace position of civil society in disaster management has been expressed extensively in the literature [21,45,80,103–106]. Because the resources and opportunities that states have are more than those of NGOs, the value of NGOs, which are one of the most important mechanisms where social capital and civil society come together, is better understood in extraordinary situations caused by large-scale disasters. NGOs, which especially have expert and qualified teams in their fields, reduce the burden on public institutions and provide contribution and practicality to disaster management.

When the geography of disaster management is considered, it is understood that public institutions make a lot of reference to the aid that can come from the surrounding provinces in the event of a possible disaster in Balıkesir. The basis of these evaluations is the understanding of “*Near is my shirt but nearer is my skin*”. A significant portion of the participants frequently referred to this understanding, stating that it would be difficult for them to help the public after a large-scale disaster in Balıkesir and that the aid would come from the surrounding provinces. However, there is a different picture in the February 6, 2023, Kahramanmaraş earthquakes. Because many provinces in the vicinity were directly and greatly affected by the earthquake. Following this, the inadequacy of the disaster management stakeholders operating in the nearby provinces emerged. For this reason, teams from different parts of Türkiye and the world carried out work in the disaster area. Therefore, it can be stated that these theses of the participants have some deficiencies. The total mental change observed after some large-scale disasters experienced in countries with strong disaster management systems such as Japan, the USA, South Korea and Bangladesh was also seen in Türkiye after the two major earthquakes in 1999. However, it is debatable whether this change is at a sufficient level. The major problems that have emerged in the disasters of the last few years have reinforced these discussions. In this respect, the positions of NGOs in disaster management and the connection between the state and NGOs are not sufficient. For example, in the TAMP, NGOs are shown as one of the important parts and support providers of disaster management. However, many public institution representatives stated that NGOs do not have any say in decision-making processes regarding disaster management and that they generally cannot establish healthy relationships with NGOs. However, the weakness of the connection between the relevant parties is relatively true, especially in the eyes of AFAD Balıkesir and BBB Fire and Disaster Management, the two most important institutions in Balıkesir disaster management. Because these two institutions stated that they established positive relations with NGOs that they thought were of high quality and contributed to disaster management. Therefore, it can be said that state-NGO relations are also affected by the character of NGOs. However, when the views of some participants are taken into consideration, it should not be ignored that the implementation dimension of TAMP contains problems. Although the understanding that civil society has a significant role in disaster management is emphasized in TAMP, the majority of participants have stated the opposite. This situation is not compatible with DRR, disaster (risk) governance, and consequently, the bottom-up approach to disaster management. Unlike these three approaches, which advocate for a community-based disaster management model that mandates the participation of civil society in decision-making processes to build a disaster-resilient and effective society while incorporating local-scale knowledge, the findings of this study reveal that NGOs have limited roles and collaboration with the state [29,39,43,45].

The two major earthquakes in Kahramanmaraş on February 6, 2023, once again revealed the disaster management problems in Türkiye. However, the importance of the role of NGOs in disasters was also understood again after the earthquakes. Both national and international NGOs took part in the processes such as search and rescue, humanitarian aid, and psychological support that started after the disaster and some of which still continue. The 911 Search and Rescue Association, DAK, and GÖNDAK located in Balıkesir provided support for search and rescue efforts after the earthquake and rescued many people alive from under the rubble. These three NGOs have another common aspect besides search and rescue. It is understood from the interviews that a significant portion of public institutions highlighted the quality of these three NGOs and evaluated them as positive examples. In addition, when their opportunities, disaster activities, perspectives on disaster management, visions, and projects are evaluated, it can be argued that these NGOs are ahead of other NGOs in Balıkesir in terms of quality level. Therefore, it is not a coincidence that the aforementioned NGOs took part in the search and rescue operations carried out after one of the biggest disasters Türkiye has ever faced.

As previously mentioned, in some developed and even developing countries, the roles of NGOs in disaster management plans and strategies are not merely theoretical but are also put into practice. Australia is a striking example in this regard. Although civil society in the country is not entirely free and operates under strict control, the NSDR, which defines the country’s disaster management framework, has granted civil society the recognition it deserves. Consequently, the opinions of civil society have become highly significant in disaster management projects. The fact that an NGO leads the “*Be Ready Warrandyte*” project, which was established to combat wildfires—one of the most potentially impactful disasters in the country—demonstrates the importance attributed to civil society [19,78,79]. However, in the case of Balıkesir, the extent to which NGOs’ opinions are considered and, consequently, the value

attributed to them remains debatable. The “*Provincial Disaster Risk Reduction Plan (İRAP)*” prepared in cooperation with the Balıkesir Governorship and AFAD Balıkesir constitutes one of the important indicators that reveal the position of NGOs in Balıkesir disaster management. The plan emphasizes that NGOs, together with public institutions, local governments, organized industrial zone directorates, professional chambers, universities and large-scale industrial organizations, contributed to the preparation process of İRAP and TAMP Balıkesir. At the same time, the support of NGOs in all kinds of disaster drills carried out throughout the province is underlined [107]. Although these emphases are important, they do not provide a clear idea of how much the perspectives of NGOs are reflected in the plans. Indeed, while participants stated that they value volunteerism and do not exclude civil society when discussing their relationships with NGOs, they also argued that the primary actor in disaster management is public institutions. This further reinforces reasonable doubts about a ‘*love for civil society*’ that remains confined to legal frameworks.

Informal volunteers, who emerge due to motivations such as social movements or social responsibility, hold a crucial position in the disaster management system, as emphasized by Nahkur et al. [12]. In large-scale and complex disasters where NGOs cannot organize effectively, the prompt response of informal volunteers can be extremely vital. In this context, during the chaotic post-disaster period, informal volunteers contribute significantly by conveying the positions, conditions, and needs of disaster victims to the relevant authorities, as well as by raising public awareness. Thus, informal volunteers not only provide direct assistance to disaster victims but also help guide public institutions and NGOs to the right locations. Social media plays a major role in this coordination. For instance, during the February 6, 2023, Kahramanmaraş earthquakes in Türkiye, some well-known figures, including artists, athletes, and opinion leaders, shared real-time updates about the affected region via social media, facilitating the rapid deployment of aid teams to critical areas. Furthermore, some disaster victims trapped under the rubble reported their locations through social media, enabling their rescue.

In terms of the position of NGOs in disaster management and their ability to take responsibility, Japan and the USA are at the forefront. For example, VOAD in the USA constitute one of the most critical partners of public institutions in local disaster management. The freedom to take initiative in post-disaster work is a major factor in VOAD’s success [71]. However, most of the participants defended the superiority of AFAD in the hierarchy and the fact that NGOs do not have the ability to assume responsibility. Although the “*Paternal State*” mentality that has been going on in Türkiye for many years is related to this situation, it is not the only reason. Because, as confirmed by the representatives of public institutions, NGOs have problems in establishing order unless they have a leader to guide them. The İzmir Earthquake example was frequently used by the participants to present the problem. Those who defend this view believe that acting outside the system organized by the state causes some problems. Ultimately, when examined in depth, the implementation shortcomings of TAMP, along with the internal and external barriers faced by NGOs, reveal that disaster governance in Balıkesir is not functioning effectively. Consequently, there are deficiencies in both sustainable disaster management and development.

6. Recommendations

In Türkiye, the official status of NGOs in disaster management can be considered relatively ineffective. Although NGOs are most prominently included in TAMP, there are still issues due to implementation deficiencies. From this perspective, the impact of NGOs in Bangladesh, a developing country like Türkiye, is striking and could serve as a model for Türkiye. As previously mentioned, one of the institutions under the DMB responsible for shaping disaster management policies in Bangladesh is the NDMAC, where the proportion of NGOs is remarkably high. This institution plays a crucial role in building community-based disaster management [60]. Just like in Bangladesh, disasters in Türkiye should also be seen as an opportunity. In this regard, a commission should be established by NGOs under the umbrella of AFAD to provide a ‘*democratic touch*’ from civil society to disaster management, thereby taking a step toward building trust between the state and NGOs.

Following the earthquake of October 30, 2020, with a magnitude of 6.6 and an epicenter in İzmir-Seferihisar, AFAD established an accreditation system encouraging the participation of NGOs that met specific criteria in post-disaster processes. However, during the February 6, 2023, Kahramanmaraş earthquakes, issues in implementing this system became evident, as NGOs and local communities that did not meet the criteria were also involved in search and rescue operations. Therefore, if the AFAD accreditation system is improved and strictly enforced through rigorous monitoring, the impact of NGOs in disaster management will become more significant. Additionally, considering that AFAD Balıkesir and BBB Fire and Disaster Management representatives have mentioned the presence of unqualified disaster-themed NGOs in Balıkesir and Türkiye, it would be beneficial not only to establish an umbrella organization similar to EMV in Nigeria [65] and Vapepa in Finland [12] but also to improve the quality of the NGOs involved in this organization through training. As a result, these aspects will create a more favorable environment for increasing the state’s trust in NGOs.

This study, which focuses on Balıkesir, naturally contains certain limitations. These limitations stem not only from the narrow spatial scale but also from potential biases in the perspectives of institutional representatives toward NGOs. Indeed, in a similar study conducted in another city in Türkiye, participants’ views on NGOs could be entirely different or similar. At this point, while generalizing the findings to the whole of Türkiye would be an ambitious approach, the study is at least likely to provide insights into the perception of NGOs in Türkiye.

Future studies examining the roles of NGOs in disaster management and the perspectives of public institutions and local communities on these roles in various provinces of Türkiye—particularly in metropolitan areas such as İstanbul, İzmir, Ankara, and Bursa, given their potential impact—would make significant contributions to the literature. Additionally, conducting longitudinal studies to analyze the perceptions of public institutions and local communities, which are likely to change over time, is also of great importance.

Although the scope and characteristics of disaster management/governance systems in developed and developing countries are

important, how different political regimes and administrative mechanisms shape disaster management frameworks is also a highly significant issue. For instance, exploring the differences between disaster management systems in totalitarian and liberal regimes, as well as how these differences impact the effectiveness of disaster response, would provide a valuable perspective to the literature in future studies. Conducting such research would help clarify the legitimacy and validity of a bottom-up approach to disaster management more clearly.

CRedit authorship contribution statement

Oğuzhan Özkan: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alper Uzun:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization.

Funding disclosure

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The research was produced from Oğuzhan Özkan's doctoral thesis titled Afet Yöneti(şi)minde Sivil Toplum Kuruluşlarının Rolü: Balıkesir İli Örneği (The Role of Non-Governmental Organizations in Disaster Management/Governance: The Case of Balıkesir Province). We would like to thank all representatives of public institutions who formed the sample of the research and contributed to the study by agreeing to be interviewed. We are also grateful to Balıkesir University Social And Humanities Research Ethics Committee for approving the research.

Data availability

The data that has been used is confidential.

References

- [1] B.H. Pandey, K. Okazaki, Community-based disaster management: empowering communities to cope with disaster risks, *Reg. Dev. Dialog.* 26 (2) (2005) 1–8.
- [2] P. Palttala, C. Boano, R. Lund, M. Vos, Communication gaps in disaster management: perceptions by experts from governmental and non-governmental organizations, *J. Contingencies Crisis Manag.* 20 (1) (2012) 2–12.
- [3] P. Rai, V. Khawas, Traditional knowledge system in disaster risk reduction: exploration, acknowledgement and proposition, *Jamba, Journal of Disaster Risk Studies* 11 (1) (2019) 1–7.
- [4] J. Dostál, Lessons of cooperation between government and non-governmental organizations in emergency management in the Czech Republic, *International Journal of Safety and Security Engineering* 5 (3) (2015) 203–221.
- [5] Y. Acer, D. Okumuş, Türkiye'nin uluslararası yardım çağrısı ve karşılığı, *kriter dergi*. <https://kriterdergi.com/dosya-deprem-ve-afet-yonetimi/turkiyenin-uluslararasi-yardim-cagrisi-ve-karsiligi>, 2023. (Accessed 12 October 2024).
- [6] Deprem sonrası uluslararası yardımlar ve işbirliği türk diplomasisini nasıl etkileyecek? - BBC News Türkçe, (n.d.). <https://www.bbc.com/turkce/articles/c3ggn8epzg8o> (accessed October 12, 2024).
- [7] 6 soruda 6 Şubat depremleri - BBC news Türkçe, (n.d.). <https://www.bbc.com/turkce/articles/cnen1eep0e1o> (accessed October 12, 2024).
- [8] L. Drennan, L. Morrissey, Resilience policy in practice — surveying the role of community based organisations in local disaster management, *Local Disaster Management* (2020) 36–57.
- [9] E. Dube, Improving disaster risk reduction capacity of district civil protection units in managing veld fires: a case of Mangwe district in Matabeleland South province, Zimbabwe, *Jambá, Journal of Disaster Risk Studies* 7 (1) (2015) 1–13.
- [10] T. Fitzpatrick, J. Molloy, The role of NGOs in building sustainable community resilience, *International Journal of Disaster Resilience in the Built Environment* 5 (3) (2014) 292–304.
- [11] D. Mondal, S. Chowdhury, D. Basu, Role of non governmental organization in disaster management, *Research Journal of Agricultural Sciences* 6 (2) (2015) 1485–1489.
- [12] O. Nahkur, K. Orru, et al., The engagement of informal volunteers in disaster management in Europe, *Int. J. Disaster Risk Reduct.* 83 (2022) 1–14.
- [13] E.S. Park, D.K. Yoon, The value of NGOs in disaster management and governance in South Korea and Japan, *Int. J. Disaster Risk Reduct.* 69 (2022) 1–15.
- [14] M. Rapeli, Social capital in social work disaster preparedness plans: the case of Finland, *Int. Soc. Work* 61 (6) (2018) 1054–1066.
- [15] A.A. Shah, A. Khan, A. Ullah, N.A. Khan, B.A. Alotaibi, The role of social capital as a key player in disaster risk comprehension and dissemination: lived experience of rural communities in Pakistan, *Nat. Hazards* 120 (5) (2024) 4131–4157.
- [16] J.N. Acharya, W. Koedsin, K. Techato, Strategies for disaster management and the role of effective governance in Nepal, *J. Water Land Dev.* (2025) 130–135.
- [17] H.M.L. Hermansson, Disaster management collaboration in Turkey: assessing progress and challenges of hybrid network governance, *Public Adm.* 94 (2) (2016) 333–349.
- [18] Y. Kang, Dispersed domination through patron-clientelism: the evolution of the local state–NGO relationship in post-disaster Sichuan, *J. Contemp. China* 29 (124) (2020) 598–613.
- [19] B. McLennan, J. Whittaker, J. Handmer, The changing landscape of disaster volunteering: opportunities, responses and gaps in Australia, *Nat. Hazards* 84 (2016) 2031–2048.
- [20] C.D.B. Burt, *Managing the Public's Trust in Non-profit Organizations*, Springer Science & Business Media, Berlin, 2014.
- [21] F. Çakı, Afet yönetiminde sivil toplum ve Güven Algıları, *Sosyal ve Kültürel Araştırmalar Dergisi (SKAD)* 6 (2020) 1–45.

- [22] W.W. Weng, C.K. Woo, Y.S. Cheng, T. Ho, I. Horowitz, Public trust and corruption perception: disaster relief, *Appl. Econ.* 47 (46) (2015) 4967–4981.
- [23] G. Bankoff, D. Hilhorst, The politics of risk in the Philippines: comparing state and NGO perceptions of disaster management, *Disasters* 33 (4) (2009) 686–704.
- [24] T.C. İçişleri, Bakanlık afet ve acil durum Yönetimi Başkanlığı - afet İstatistikleri, n.d. <https://www.afad.gov.tr/afet-istatistikleri>. (Accessed 2 April 2025).
- [25] Public EM-DAT platform, n.d. <https://public.emdat.be/data>. (Accessed 2 April 2025).
- [26] Countries with the Most Natural Disasters, Statista, 2023 n.d. <https://www.statista.com/statistics/269652/countries-with-the-most-natural-disasters/>. (Accessed 2 April 2025).
- [27] INFORM report 2024 - Publications office of the EU, n.d. <https://op.europa.eu/en/publication-detail/-/publication/b0ac7f32-e4f1-11ee-8b2b-01aa75ed71a1/language-en>. (Accessed 2 April 2025).
- [28] D. Paton, Disaster risk reduction: psychological perspectives on preparedness, *Aust. J. Psychol.* 71 (4) (2019) 327–341.
- [29] J. Twigg, Disaster Risk Reduction, Humanitarian Policy Group, England, 2015.
- [30] R.E. Kaspersen, A.M. Berberian, Integrating Science and Policy: Vulnerability and Resilience in Global Environmental Change, Earthscan, New York, 2011.
- [31] J.C. Gaillard, J. Mercer, From knowledge to action: bridging gaps in disaster risk reduction, *Prog. Hum. Geogr.* 37 (1) (2013) 93–114.
- [32] I. Kelman, Climate change and the sendai framework for disaster risk reduction, *International Journal of Disaster Risk Science* 6 (2015) 117–127.
- [33] J. Weichselgartner, P. Pigeon, The role of knowledge in disaster risk reduction, *International Journal of Disaster Risk Science* 6 (2015) 107–116.
- [34] T. Izumi, R. Shaw, R. Djalante, M. Ishiwatari, T. Komino, Disaster risk reduction and innovations, *Progress in Disaster Science* 2 (2019) 1–8.
- [35] A. Aitsi-Selmi, V. Murray, et al., Reflections on a science and technology agenda for 21st century disaster risk reduction: based on the scientific content of the 2016 UNISDR science and technology conference on the implementation of the sendai framework for disaster risk reduction 2015–2030, *International Journal of Disaster Risk Science* 7 (2016) 1–29.
- [36] J. Weichselgartner, R. Kaspersen, Barriers in the science-policy-practice interface: toward a knowledge-action-system in global environmental change research, *Glob. Environ. Change* 20 (2) (2010) 266–277.
- [37] H.D. Cash, J.W. Hoyle, A.J. Sutton, Development under extreme conditions: forensic bioinformatics in the wake of the world trade center disaster, *Biocomputing* 2003 (2002) 638–653.
- [38] A.W. Whelchel, B.G. Reguero, B. van Wesenbeeck, F.G. Renaud, Advancing disaster risk reduction through the integration of science, design, and policy into eco-engineering and several global resource management processes, *Int. J. Disaster Risk Reduct.* 32 (2018) 29–41.
- [39] D. Alexander, The Meaning of Disaster: a Reply to Wolf Dombrowsky, Xlibris Press, The USA, 2005.
- [40] J.W. Ammann, S. Dannenmann, L. Vulliet, RISK21 - Coping with risks due to natural hazards in the 21st century. Proceedings of the RISK21 Workshop, CRC Press, UK, 2006.
- [41] UNISDR, Towards the Post-2015 Framework for Disaster Risk Reduction Tackling Future Risks, Economic Losses and Exposure, United Nations, USA, 2013.
- [42] M. Gall, S.L. Cutter, K.H. Nguyen, Governance in Disaster Risk Management, Integrated Research on Disaster Risk, China, 2014.
- [43] K. Tierney, Disaster governance: social, political, and economic dimensions, *Annu. Rev. Environ. Resour.* 37 (2012) 341–363, <https://doi.org/10.1146/ANNUREV-ENVIRON-020911-095618/CITE/REFWORKS>.
- [44] D. Alexander, An interpretation of disaster in terms of changes in culture, society and international relations, in: R.W. Perry, E.L. Quarantelli (Eds.), What is a Disaster? New Answers to Old Questions, Xlibris Press, USA, 2005, pp. 25–38.
- [45] V. Sandoval, M. Voss, Disaster Governance and Vulnerability: the Case of Chile, vol. 4, 2016, pp. 107–116, <https://doi.org/10.17645/pag.v4i4.743>.
- [46] M. Voss, The vulnerable can't speak. An integrative vulnerability approach to disaster and climate change research, *Behemoth: A Journal on Civilisation* 3 (2008) 39–56, <https://doi.org/10.1524/behe.2008.0022>.
- [47] B. Wisner, I. Kelman, T. Monk, J. Bothara, D. Alexander, D. Dixit, M. Petal, School seismic safety: falling between the cracks, *Earthquake* 3 (2004) 1–56.
- [48] M. Kahn, The death toll from natural disasters: the role of income, geography, and institutions, *Rev. Econ. Stat.* 87 (2005) 271–284.
- [49] P. Peduzzi, H. Dao, C. Herold, F. Mouton, Assessing global exposure and vulnerability towards natural hazards: the disaster risk index, *Nat. Hazards Earth Syst. Sci.* 9 (2009) 1149–1159.
- [50] P.A. Raschky, Institutions and the losses from natural disasters, *Nat. Hazards Earth Syst. Sci.* 8 (2008) 627–634.
- [51] L. Stough, Disaster and Social Vulnerability: the Case of Undocumented Mexican Migrant Workers, Edwin Mellen Press, USA, 2010.
- [52] A.G. Nick, E. Savoia, L. Elqura, S. Crowther, B. Cohen, M. Leary, T. Wright, J. Auerbach, H.K. Koh, Emergency preparedness for vulnerable populations: people with special health-care needs, *Publ. Health Rep.* 124 (2009) 338–343.
- [53] E. Enarson, Women Confronting Natural Disaster: from Vulnerability to Resilience, Lynne Rienner Publication, USA, 2012.
- [54] Z.Y. Odabaş, Sürdürülebilir Afet Yönetimi ve Kadın, Ankara Üniversitesi Rektörlüğü Yayınları, Türkiye, 2010.
- [55] C. Pincha, Indian Ocean Tsunami Through the Gender Lens: Insights from Tamil Nadu, India. Earthworm Books, India, 2008.
- [56] N. Bayraktar, Ü.D. Yılmaz, Vulnerability of elderly people in disasters: a systematic review, *Türk. J. Gastroenterol.* 21 (2018) 467–482.
- [57] A. Bodstein, V.V.A. Lima, A. Barros, The vulnerability of the elderly in disasters: the need for an effective resilience policy, *Ambiente Sociedade* 17 (2014) 157–174.
- [58] J. Hartog, Disaster Resilience in an Ageing World: How to Make Policies and Programmes Inclusive of Older People, HelpAge International, UK, 2014.
- [59] GIVE2ASIA, IIRR, Building Community Resilience: Mapping the Journey of Local Community Based Ngos in Developing Sustainable Preparedness Programs, White Paper, USA, 2017.
- [60] A.K.M.A. Sabur, Disaster Management System in Bangladesh, *India Q.* vol. 68, 2012, pp. 29–47.
- [61] Nigeria: Humanitarian overview (as of 31 October 2014) | OCHA, (n.d.). <https://www.unocha.org/publications/report/nigeria/nigeria-humanitarian-overview-31-october-2014> (accessed April 2, 2025).
- [62] Disaster preparedness & management – office of the prime minister – a coordinated, Responsive and Accountable Government for Socio-Economic Transformation, (n.d.). <https://opm.go.gov/disaster-preparedness-management/> (accessed April 2, 2025).
- [63] R. Ariyabandu, Water Policy Development in a multi-party System of Governance : a Case Study of Sri Lanka 1, India, Routledge, 2018.
- [64] K.C. Dipendra, Determinants of perceived innovation in non-governmental organizations in Nepal, *Nonprof. Manag. Leader.* 34 (3) (2024) 683–705.
- [65] NEMA Nigeria – official site, (n.d.). <https://www.nema.gov.ng/> (accessed April 2, 2025).
- [66] B. Rohwerder, Crisis management models in Africa, Helpdesk Research Report.
- [67] About us | FEMA.gov, (n.d.). <https://www.fema.gov/about> (accessed October 13, 2024).
- [68] L.K. Comfort, Risk, security disaster and management, *Annu. Rev. Polit. Sci.* 8 (1) (2005) 335–356.
- [69] L.K. Comfort, Managing intergovernmental responses to terrorism and other extreme events, *Publius* 32 (4) (2002) 29–50.
- [70] G. Haddow, J.A. Bullock, K. Haddow, Global Warming, Natural Hazards, and Emergency Management, Routledge, New York, 2017.
- [71] D.-Y. Jung, K.M. Ha, A comparison of the role of voluntary organizations in disaster management, *Sustainability* 13 (4) (2021) 1–15.
- [72] D.F. Lorenz, K. Schulze, M. Voss, Emerging citizen responses to disasters in Germany. Disaster myths as an impediment for a collaboration of unaffiliated responders and professional rescue forces, *J. Contingencies Crisis Manag.* 26 (2018) 358–367.
- [73] T. Bodland, M. Granberg, The community resilience challenge—A case study from Sweden, Reframing Urban Resilience Implementation Conference Proceedings – Long Paper, 1–15.
- [74] MSB – the Swedish civil contingencies agency | MSB, (n.d.). <https://www.msb.se/en/> (accessed April 2, 2025).
- [75] R. Johansson, E. Danielsson, L. Kvarnlöf, K. Eriksson, R. Karlsson, At the external boundary of a disaster response operation: the dynamics of volunteer inclusion, *J. Contingencies Crisis Manag.* 26 (2018) 519–529, <https://doi.org/10.1111/1468-5973.12228>.
- [76] FORF - lokal kompetanse lokal trygghet, (n.d.). <https://www.forf.no/?id=249&Rapporten%20som%20skal%20l%F8fte%20de%20frivillige%20i%2020redningstjenesten> (accessed April 2, 2025).
- [77] H. Raisio, A. Puustinen, T. Norri-Sederholm, J. Jalava “Those who agree to play on our terms will be taken in”: a qualitative study on the perceptions of public authorities and NGO representatives regarding self-organizing fourth-sector activity, *Public Adm. Q.*, 43(3), 289–329.
- [78] COAG, (n.d.). <https://www.parliament.nsw.gov.au/researchpapers/Pages/coag.aspx> (accessed April 2, 2025).

- [79] T. Tenbenschel, J. Dwyer, J. Lavoie, How not to kill the golden goose: reconceptualizing accountability environments of third-sector organizations, *Public Manag. Rev.* 16 (2014) 925–944.
- [80] A. Özerdem, T. Jacoby, *Earthquake Relief in Japan, Turkey and India*, IB Tauris, UK, 2006.
- [81] K. Hirata, *Civil Society in Japan: the Growing Role of NGOs in Tokyo's Aid and Development Policy*, Palgrave Macmillan, New York, 2002.
- [82] L. Salamon, H. Anheier, R. List, et al., *La Sociedad Civil Global: Las Dimensiones Del Sector No Lucrativo*, Fundación BBVA, Bilbao, 1999.
- [83] Japan: Disaster countermeasures basic act (Act no. 223 of 15 November 1961; revised June 1997) | PreventionWeb, (n.d.). <https://www.preventionweb.net/publication/japan-disaster-countermeasures-basic-act-act-no-223-15-november-1961-revised-june-1997> (accessed April 2, 2025).
- [84] B. Dollery, Y. Kinoshita, K. Yamazaki, Humanitarian co-production in local government: the case of natural disaster volunteering in Japan, *Local Gov. Stud.* 46 (6) (2020) 959–978.
- [85] Ç. Akgüngör, Toplumbilim perspektifinden afeti düşünmek, *LAÜ Sosyal Bilimler Dergisi* 1 (1) (2010) 3–22.
- [86] R. Jalali, Civil society and the state: turkey after the earthquake, *Disasters* 26 (2) (2002) 120–139.
- [87] H. Yavaş, Türkiye'de doğal afetlerin merkez-yerel ilişkiler açısından yönetim sorunları, *Dokuz Eylül Üniversitesi Sosyal Bilimler Enstitüsü Dergisi* 7 (3) (2005) 280–301.
- [88] S. Schneider, *Flirting with Disaster: Public Management in Crisis Situations*, ME Sharpe, New York, 1995.
- [89] T.A. Birkland, Disasters, lessons learned, and fantasy documents, *J. Contingencies Crisis Manag.* 17 (2009) 146–156.
- [90] A. Boin, P. T Hart, A. McConnell, Crisis exploitation: political and policy impacts of framing contests, *J. Eur. Publ. Pol.* 16 (2009) 81–106.
- [91] T. Aydnır, H. Özgür, Yönetişim ve doğal afetler: Türkiye'de doğal Afet Yönetişimi Ne kadar ve Hangi Biçimde Mümkün? Aydın, Adnan Menderes Üniversitesi (2014).
- [92] T.C. İçişleri Bakanlığı afet ve acil durum Yönetimi Başkanlığı - türkiye afet müdahale planı, (n.d.) <https://www.afad.gov.tr/turkiye-afet-mudahale-planı> (accessed April 3, 2025).
- [93] T. Erkal, M. Değerliyurt, Türkiye'de afet yönetimi, *Doğu Coğrafya Dergisi* 14 (22) (2009) 147–164.
- [94] C. Benson, M. Myers, J. Twigg, NGO initiatives in risk reduction: an overview, *Disasters* 25 (2001) 199–215.
- [95] W.N. Carter, *Disaster Management Handbook*, Asian Development Bank, The Philippines, 2008.
- [96] U. Kuckartz, S. Rädiker, *Analyzing Qualitative Data with MAXQDA*, Springer International Publishing, The USA, 2019.
- [97] J. Saldaña, *Coding Techniques for Quantitative and Mixed Data*, Routledge, The USA, 2019.
- [98] Yokohama strategy and plan of action for a safer world: guidelines for natural disaster prevention, preparedness and mitigation | UNDRR, (n.d.). <https://www.undrr.org/publication/yokohama-strategy-and-plan-action-safer-world-guidelines-natural-disaster-prevention> (accessed October 13, 2024).
- [99] Guidelines for aid agencies on disaster mitigation | IUCN library system, (n.d.). <https://portals.iucn.org/library/node/20508> (accessed October 13, 2024).
- [100] R. Shaw, The role of nongovernmental organizations in earthquake disaster management: an Asian perspective, *Reg. Dev. Dialog.* 24 (1) (2013) 117–119.
- [101] M. Castells, A rejoinder: on power, identities and culture in the network society, *New Polit. Econ.* 3 (3) (1998) 473–483.
- [102] M.L. Hall, A.C.K. Lee, C. Cartwright, S. Maharatta, J. Karki, P. Simkhada, The 2015 Nepal earthquake disaster: lessons learned one year on, *Public Health* 145 (2017) 39–44.
- [103] E. Coles, P. Buckle, Developing community resilience as a foundation for effective disaster recovery, *Aust. J. Emerg. Manag.* 19 (4) (2004) 6–15.
- [104] N. Kapucu, V. Garayev, Collaborative decision-making in emergency and disaster management, *Int. J. Publ. Adm.* 34 (6) (2011) 366–375.
- [105] L. Pearce, Disaster management and community planning, and public participation: how to achieve sustainable hazard mitigation, *Nat. Hazards* 28 (2003) 211–228.
- [106] W. Smith, J. Dowell, A case study of co-ordinative decision-making in disaster management, *Ergonomics (Lond.)* 43 (8) (2000) 1153–1166.
- [107] İrap-Ek, (n.d.). <https://balikesir.afad.gov.tr/irap-ek> (accessed October 13, 2024).
- [108] Teşkilat şeması. <https://www.afad.gov.tr/teskilat-semasi>, 2024. (Accessed 13 October 2024).